

Reading Notes on Framed Cobordisms

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Preface

These are my reading notes and final report for the Spring 2026 reading course on differential topology and cobordism theory. The reading mainly covered the content of *Topology from Differential Viewpoint* by John Milnor and supplemented by Daniel S. Freed's lecture notes *Bordism: Old and New*.

1 Orientation of Manifolds

The study of geometry and topology always starts from some linear algebra :

Orientation of Vector Spaces. As claimed above, almost every structure on a manifold is defined pointwise on tangent spaces (or, in other words, defined as the section of some bundle). Thus, we may start with the orientation on vector spaces: Let V be an n -dimensional vector space (over $\mathbb{R}$), we define the set of k -ordered frames over V as

$$\text{Fr}_k(V) := \{(v_1, \dots, v_k) \in V^{\times k} \mid v_1, \dots, v_k \text{ linearly independent}\}$$

In particular, an n -frame in $\text{Fr}_n(V)$ is an ordered basis of V and each n -frame corresponds to an element in the exterior algebra $\Lambda^n V$. Note that there is a natural $\text{GL}^+(n, \mathbb{R})$ action on $\text{Fr}_n(V)$ by the change of basis matrix $[\text{id}_V]_{\beta_1}^{\beta_2}$ for $\beta_1, \beta_2 \in \text{Fr}_n(V)$.

Definition 1.1 (Orientation of Vector Spaces). An orientation over a n -dimensional vector space V is given by the quotient of the n -frame over V by the $\text{GL}^+(n, \mathbb{R})$ action

$$\text{Or}(V) := \text{Fr}_n(V) / \text{GL}^+(n, \mathbb{R})$$

An orientation on V is an element $o \in \text{Or}(V)$, and a frame is called positively oriented with respect to o if its $\text{GL}^+(n, \mathbb{R})$ orbit is o .

It is not hard to show that every vector space has exactly two orientations by the following proposition

Proposition 1.2. Let f_1 and f_2 in $\Lambda^n V$ be element in the exterior algebra over V corresponding to n -frames $\beta_1, \beta_2 \in \text{Fr}_n(V)$, $\dim V = n$. Then $\beta_1 \sim \beta_2$ if and only if $\exists c > 0 : f_1 = cf_2$.

Proof. Recall that the top dimension exterior algebra is given by

$$\Lambda^n V := V^{\otimes n} / I, \quad I := \langle v \otimes v \in V^{\otimes 2} \mid \forall v \in V \rangle$$

and for n -frame, there is natural correspondence $\text{Fr}_n(V) \hookrightarrow V^{\otimes n} \rightarrow \Lambda^n V \setminus \{0\}$ which sends

$$(v_1, \dots, v_n) \mapsto v_1 \wedge \dots \wedge v_n \neq 0$$

The image is nonzero by the linear independence of a frame.

($\Rightarrow$) For n -frames $\beta_1 = (v_1, \dots, v_n)$ and $\beta_2 = (w_1, \dots, w_n)$ such that $\beta_1 \sim \beta_2$, let

$$f_1 = v_1 \wedge \dots \wedge v_n, \quad f_2 = w_1 \wedge \dots \wedge w_n$$

since $\dim V = n$, the top degree exterior algebra is $\binom{n}{n} = 1$. Consider the unique linear isomorphism $A \in \text{GL}^+(n, \mathbb{R})$ such that $A(w_i) = v_i$. Then, the isomorphism induces the isomorphism on exterior algebra by

$$(\Lambda^n A)(f_2) = A(w_1) \wedge \dots \wedge A(w_n) = v_1 \wedge \dots \wedge v_n = f_1$$

But by the multilinearity of the wedge product, this is just saying

$$f_1 = \det(A) \cdot f_2$$

since $A \in \mathrm{GL}^+(n, \mathbb{R})$, we know that $\det A > 0$, which proves the claim.

($\Leftarrow$) The proof of this direction is the same as the previous statement, since for any two bases β_1, β_2 of V in $\mathrm{Fr}_n(V)$, one can find the unique linear isomorphism A that sends β_1 to β_2 . The claim of the induced linear map above has shown that $\det A > 0$, which means $A \in \mathrm{GL}^+(n, \mathbb{R})$, which means f_1 and f_2 are in the same orbit. $\square$

Thus, orientation can also be described by the ray in the top order exterior algebra $\mathbb{R}_{>0} \cdot f \subset \Lambda^n V \setminus \{0\}$. Another way to describe this fact is to use the determinant map, as we have shown in the proof of Proposition 1.2, the determinant

$$\det : \mathrm{GL}(n, \mathbb{R}) \rightarrow \mathbb{R}^\times$$

can be view as the map $\mathrm{GL}(n, \mathbb{R}) \times \Lambda^n V \rightarrow \Lambda^n V$ given by $A \cdot f = \det(A)f$. In particular, the exterior algebra satisfies the following property

Proposition 1.3. *The short exact sequence of finite-dimensional real vector spaces*

$$0 \longrightarrow V_1 \xrightarrow{i} V \xrightleftharpoons[\pi]{s} V_2 \longrightarrow 0$$

the chosen split $s : V_2 \rightarrow V$ leads to a isomorphism of exterior algebra

$$\Lambda^* V_1 \otimes \Lambda^* V_2 \xrightarrow{\cong} \Lambda V$$

where the left-hand side is the graded tensor product, the degree of which satisfies $|a \otimes b| = |a| + |b|$.

In other words, let V_1 and V_2 be vector spaces with dimensions n_1 and n_2 , denote their direct sum as $V := V_1 \oplus V_2$. Then the exterior algebra satisfies

$$\Lambda^k V \cong \bigoplus_{a+b=k} \Lambda^a V_1 \otimes \Lambda^b V_2$$

In particular, the isomorphism top degree part is canonical, given by $\Lambda^{n_1+n_2} V \cong \Lambda^{n_1} V_1 \otimes \Lambda^{n_2} V_2$.

Proof. To prove the statement, we may simply construct the isomorphism from the short exact sequence. Given the section of a short exact sequence $s : V_2 \rightarrow V$ such that $\pi \circ s = \mathrm{id}_{V_2}$. Then, by exactness of the sequence, given any $v \in V$ consider $v - (s \circ \pi)(v)$, since

$$\pi(v - (s \circ \pi)(v)) = \pi(v) - (\pi \circ s \circ \pi)(v) = 0$$

$v - (s \circ \pi)(v) \in \ker(\pi) = i(V_1)$. Thus, there exists $v_1 \in V_1$ such that

$$v = i(v_1) + (s \circ \pi)(v)$$

On the other hand, if there is a nonzero vector $w \in i(V_1) \cap s(V_2)$, then by exactness, $\pi(w) = (\pi \circ i)(w_1) = 0$ for some $w_1 \in V_1$. Since $\pi \circ s = \mathrm{id}_{V_2}$, $s : V_2 \rightarrow V$ is injective and thus $w = s(0) = 0$. This shows that $i(V_1) \cap s(V_2) = \{0\}$, which means

$$V = i(V_1) \oplus s(V_2)$$

Then, let $A := \Lambda^* V_1 \otimes \Lambda^* V_2$, with the canonical inclusion $i : V_1 \hookrightarrow V$ and $s : V_2 \hookrightarrow V$, then the induced map is given by

$$\Phi_s : A \rightarrow \Lambda^*(V_1 \oplus V_2), \quad \Phi_s(\alpha \otimes \beta) = \Lambda^* i(\alpha) \wedge \Lambda^* s(\beta)$$

It remains to prove that Φ_s is indeed a graded algebra isomorphism. Since the maps Λ^*i and Λ^*s are induced homomorphisms, it is sufficient to just give an inverse to Φ_s . Define the linear map

$$j_s : V \rightarrow V_1 \oplus V_2 \cong A^1, \quad j_s(v) = \text{proj}_1(v) \otimes 1 + 1 \otimes \text{proj}_2(v)$$

Where the projection $\text{proj}_i : V \rightarrow V_i$ is just the natural projection due to the direct sum, since the degree 1 part $A^1 = (\Lambda^1 V_1 \otimes \mathbb{R}) \oplus (\mathbb{R} \otimes \Lambda^1 V_2) \cong V_1 \oplus V_2$. Due to the universal property of exterior algebra, since the image map j_s is the degree 1 part of A , it satisfies $j_s(v)^2 = 0$. Thus, the homomorphism can be uniquely extended to a graded algebra homomorphism

$$\Psi_s : \Lambda^* V \rightarrow A, \quad \Psi_s|_{\Lambda^1 V} = j_s$$

Explicitly, the map is defined by $\Psi_s(v) = \text{proj}_1(v) \otimes 1 + 1 \otimes \text{proj}_2(v)$. We shall check that $\Psi_s = \Phi_s^{-1}$:

1. For any $v_1 \in V_1$, $\Psi_s(i(v_1)) = \text{proj}_1(i(v_1)) \otimes 1 + 1 \otimes \text{proj}_2(i(v_1)) = v_1 \otimes 1$.
2. For any $v_2 \in V_2$, $\Psi_s(s(v_2)) = \text{proj}_1(s(v_2)) \otimes 1 + 1 \otimes \text{proj}_2(s(v_2)) = 1 \otimes v_2$.

Then, for any generator of $V_1 \cong \Lambda^1 V_1 \otimes \mathbb{R}$ and $V_2 \cong \mathbb{R} \otimes \Lambda^1 V_2$, $\Psi_s \circ \Phi_s$ is identity. Since A is generated by the generators of $\Lambda^1 V_1 \otimes \mathbb{R}$ and $\mathbb{R} \otimes \Lambda^1 V_2$, $\Psi_s \circ \Phi_s$,

$$\Psi_s \circ \Phi_s = \text{id}_A$$

Conversely, for any $v \in V$, one has

$$\begin{aligned} \Phi_s \circ \Psi_s(v) &= \Phi_s(\text{proj}_1(v) \otimes 1 + 1 \otimes \text{proj}_2(v)) \\ &= i(\text{proj}_1(v)) \wedge 1 + 1 \wedge s(\text{proj}_2(v)) = v \end{aligned}$$

Since $\Lambda^* V$ is generated by the elements in V , $\Phi_s \circ \Psi_s = \text{id}_{\Lambda^* V}$. Thus, $\Psi_s = \Phi_s^{-1}$, which completes the proof of the proposition. $\square$

In particular, for the top degree parts, let $n_i := \dim V_i$ for $i = 1, 2$ and $n = n_1 + n_2$. The short exact sequence naturally gives the canonical isomorphism

$$\Delta : \Lambda^{n_1} V_1 \otimes \Lambda^{n_2} V_2 \xrightarrow{\cong} \Lambda^n V, \quad \Delta(\xi \otimes \eta) = \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} s(\eta)$$

since the change of section between $s, s' : V_2 \rightarrow V_2$ will only differ by $\ker(\pi) = \text{im}(i)$, and then one can find linear map $T : V_2 \rightarrow V_1$ such that $s' = s + i \circ T$ and the top degree isomorphism is given by

$$\Delta(\xi \otimes \eta) = \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} s'(\eta) = \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} s(\eta) + \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} (i \circ A)(\eta)$$

where, by the definition of wedge product, the second term is zero. Based on Proposition 1.3, one can claim that

Corollary 1.4. *With given orientation $o_i \in \text{Or}(V_i)$, the frame $(e_1, e_2) \in \text{Fr}_n(V_1 \oplus V_2)$ is positively oriented if and only if e_i is positively oriented in V_i .*

Also, in $V_1 \oplus V_2$, exchanging the order of the basis will lead to a Koszul sign

$$(e_2, e_1) = (-1)^{n_1 n_2} (e_1, e_2)$$

which follows from the natural grading on $\Lambda^n V$.

Another natural result that has been widely used in the discussion of orientation is that

Proposition 1.5. *The orientation on vector spaces V leads to a natural orientation on the dual space V^* .*

In the finite-dimensional case, this is best understood through the canonical determinant-line isomorphism $\Lambda^n(V^*) \cong (\Lambda^n V)^*$. An orientation of V^* is a choice of positive ray on $\Lambda^n(V^*)$, and the dual positive ray $(\Lambda^n V)^*$ corresponds canonically to the positive ray in $\Lambda^n V$ since $\dim \Lambda^n V = 1$. This definition is independent of the choice of a positively oriented basis. In the discussion of smooth manifold orientations, we often use the language of dual space instead of the tangent space.

Orientation of Manifolds. As for the smooth manifolds, recall that the tangent space at a point $p \in M$ is defined as

$$T_p M := \{X : \mathcal{O}_{M,p} \rightarrow \mathbb{R} \mid X \text{ is point derivation at } p\}$$

where $\mathcal{O}_{M,p} := C_{M,p}^\infty$ is the $\mathbb{R}$ -algebra of smooth function germs at $p \in M$. We may show that the orientation of the smooth manifold is actually given by the orientation on the tangent space at each point in M . We may first state the definition of orientation on smooth manifolds using an atlas:

Definition 1.6 (Orientable Atlas). Let M be an n -dimensional smooth manifold. A smooth atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is orientable if for any $\alpha, \beta \in A$ the transition map

$$\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \varphi_\alpha(U_\alpha \cap U_\beta)$$

are orientable-preserving diffeomorphism, i.e., the Jacobian $\det D(\varphi_{\alpha\beta}) > 0$. A smooth manifold is orientable if and only if it admits an orientable atlas.

An orientable atlas determines an orientation on each tangent space. Indeed, if (U, φ) is a chart with local coordinate given by $\varphi = (x^1, \dots, x^n)$, then for every $p \in U$, the tangent space $T_p M$ admits a coordinate induced (ordered) basis

$$\left(\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right)$$

and we may declare this ordered basis to be positively oriented. If $p \in U_\alpha \cap U_\beta$, then the two coordinate bases are related by the Jacobian matrix of the transition map. Since the transition map has a positive Jacobian determinant, the two coordinate bases define the same orientation of $T_p M$. Hence, an orientable atlas gives a well-defined orientation on every tangent space $T_p M$. This leads to the following proposition

Theorem 1.7. *The n -dimensional smooth manifold M is orientable if and only if the determinant line bundle $\Lambda^n TM$ admits a non-vanishing smooth section.*

Proof. ($\Rightarrow$) The key in this direction is that in the definition of the oriented atlas, we have required the transition map to be orientation preserving, i.e., for charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) with coordinates given by $\varphi_\alpha = (x^1, \dots, x^n)$ and $\varphi_\beta = (y^1, \dots, y^n)$

$$g_{\beta\alpha} := \det \left(\frac{\partial \varphi_\beta}{\partial \varphi_\alpha} \right) > 0, \quad D(\varphi_\beta \circ \varphi_\alpha^{-1}) = \frac{\partial \varphi_\beta}{\partial \varphi_\alpha} := \left[\frac{\partial y^i}{\partial x^j} \right]_{1 \leq i,j \leq n}$$

On each local chart $(U_\alpha, \varphi_\alpha)$, one can define the non-vanishing constant section in $\Lambda^n TM|_{U_\alpha}$ as

$$e_\alpha(p) = \frac{\partial}{\partial x^1} \Big|_p \wedge \dots \wedge \frac{\partial}{\partial x^n} \Big|_p$$

and by the change of coordinate formula of the chart induced basis on $U_\alpha \cap U_\beta$, the positively oriented frame on U_α and U_β satisfies the compatibility condition $e_\alpha = g_{\beta\alpha}e_\beta$, which means e_α and e_β only differ by a positive function on $U_\alpha \cap U_\beta$ by the positivity of the Jacobian of transition map. Take a partition of unity $\{\rho_\alpha\}$ corresponding to the given orientable atlas such that $\text{supp}(\rho_\alpha) \subseteq U_\alpha$ and let

$$s(p) = \sum_{\alpha} \rho_\alpha(p) e_\alpha(p) = \left(\sum_{\alpha} \rho_\alpha(p) g_{\beta\alpha}(p) \right) e_\beta(p) \quad \forall p \in M$$

this is indeed a globally smooth section when we extend e_α from U_α to M using partition of unity. We may check that s is nowhere vanishing. Since on each chart, $e_p = \partial_1 \wedge \cdots \wedge \partial_n$ is non-vanishing, and by the definition of partition of unity and the fact that $g_{\alpha\beta} > 0$, the coefficient is strictly positive. Thus s is a smooth non-vanishing section.

($\Leftarrow$) Suppose there is a non-vanishing section $s \in \Gamma(\Lambda^n TM)$. It is sufficient to claim that on any connected chart U_α , the section $s(p) = f_\alpha(p) e_\alpha(p)$ for some $f_\alpha \in C^\infty(U_\alpha)$ since the fiber $\Lambda^n T_p M$ is 1-dimensional generated by the basis $e_\alpha(p)$. By replace e_β using $-e_\beta$ when $f_\beta < 0$, one can always consider the case $f_\alpha(p) > 0 \quad \forall \alpha$, and indeed the non-vanishing section s gives a orientation on $T_p M$ for any $p \in M$ by defined that $(v_1, \dots, v_n)$ is positively oriented if there exists $\lambda > 0$

$$v_1 \wedge \cdots \wedge v_n = \lambda s(p)$$

Since the tangent bundle always trivializes on local charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$, the orientation above can be defined on each chart. It remains to show that the positive charts are mutually compatible. Let $s = f_\alpha e_\alpha$ for some $f_\alpha \in C^\infty(U_\alpha)^\times$ be the non-vanishing section of $\Lambda^n TM$ restricted on each chart, the compatibility condition of the top-degree exterior algebra on the tangent bundle is given by

$$f_\beta(p) = \det \left(\frac{\partial \varphi_\beta}{\partial \varphi_\alpha} \right) f_\alpha, \quad \forall p \in U_\alpha \cap U_\beta$$

since $e_\beta = \det(\partial \varphi_\alpha / \partial \varphi_\beta) e_\alpha$. By the fact that s is non-vanishing, the coefficient function must also be non-vanishing, and thus needs to be constantly positive or negative on connected charts (and one can always take the cover of connected charts on M). Thus, for the correct choice of basis

$$\det D(\varphi_\beta \circ \varphi_\alpha^{-1}) = \det \left(\frac{\partial \varphi_\beta}{\partial \varphi_\alpha} \right) = \frac{f_\beta}{f_\alpha} > 0,$$

for any $\alpha, \beta \in A$ such that $U_\alpha \cap U_\beta \neq \emptyset$, which shown that the atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is orientable. $\square$

As a result of the determinant line bundle definition of orientation on manifolds, we may remark that in most cases when studying smooth manifolds, the language of differential forms is more convenient. Due to the canonical isomorphism $(\Lambda^k TM)^* \cong \Lambda^k T^*M$, one can safely use the isomorphism (may not be canonical) between tangent and cotangent bundle to translate the orientation on the manifold to the language of n -forms: Recall that $\Omega^k(M) := \Gamma(\Lambda^k T^*M)$, let $n := \dim M$

Definition 1.8 (Volume Forms). The volume form on M is a n -form $\omega \in \Omega^n(M)$ such that $\forall p \in M$, $\omega_p \in \Lambda^n T_p^*M$ is a non-zero form.

and it is not hard to conclude the following theorem as an alternative definition of orientation using differential form:

Theorem 1.9. *M is orientable if and only if M admits a volume form.*

The proof of this theorem is simply consider the the no-where vanishing section $e \in \Gamma(\Lambda^n TM)$, since the fiber $\Lambda^n T_p M$ is 1-dimensional, one can always choose the dual section $\omega \in \Omega^n(M)$ such that pointwisely $\omega_p \in (\Lambda^n T_p M)^* \cong \Lambda^n T_p^* M$ is given by

$$\omega_p(e_p) = 1 \quad \forall p \in M$$

where the smoothness is preserved by the smoothness of e .

2 Normal Bundle and Framing

We may start with a special case. If we further consider the orientation of manifolds with boundary, then it is natural to consider the induced orientation on the boundary.

Recall that manifolds with boundary M is defined by the atlas $\{(U_\alpha, \varphi_\alpha)\}$ local model of upper half plane $\varphi : U \subseteq M \xrightarrow{\cong} V \subseteq \mathbb{H}^n$, and the boundary of M is defined by

$$\partial M = \{p \in M \mid \exists \alpha \in A : \varphi_\alpha(p) \in \partial \mathbb{H}^n\}$$

and thus, under the local coordinate $\varphi_\alpha = (x^1, \dots, x^n)$ of boundary charts, one can always choose boundary points to be points such that $x^1(p) = 0$, and the tangent space of ∂M as a subspace of the tangent space of M is given by $\forall p \in \partial M$

$$T_p(\partial M) := \text{span}\{\partial_2, \dots, \partial_n\}$$

Thus, we may define the "normal" direction (more generally, the normal bundle) of the boundary as $\nu_{\partial M} := TM|_{\partial M}/T(\partial M) = \mathbb{R} \cdot \partial_1$ which makes the following sequence exact:

$$0 \longrightarrow T(\partial M) \xhookrightarrow{i} TM|_{\partial M} \twoheadrightarrow \nu_{\partial M} \longrightarrow 0$$

With this notion of normal direction, we called the vector $n_p \in \mathbb{R}_{<0} \cdot \partial_1|_p$ the outward pointing normal vector (i.e., $(dx^1)_p(n_p) < 0$). Note that the normal bundle $\nu_{\partial M}$ is always trivial.

Proposition 2.1. *The normal bundle of the boundary $\nu_{\partial M} \cong \partial M \times \mathbb{R}$ is a trivial bundle.*

We shall prove this fact after defining the induced boundary orientation.

We may define the induced orientation of the boundary as

Definition 2.2 (Induced Boundary Orientation). Let M be an orientable smooth manifold with a chosen orientation, and let $n_p \in T_p M$ be the outward pointing normal vector at $p \in \partial M$. Then, a frame $(v_1, \dots, v_{n-1}) \in \text{Fr}_{n-1}(T_p(\partial M))$ is positively oriented if and only if $(n_p, v_1, \dots, v_{n-1}) \in \text{Fr}_n(T_p M)$ is positively oriented on M .

Since the definition used the coordinate charts, it is important to show the well-definedness:

Proposition 2.3. *The induced orientation on the boundary does not depend on the choice of boundary charts.*

Proof. The proof is simply to use the transition function. Let

$$\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} = (\varphi_{\alpha\beta}^1, \dots, \varphi_{\alpha\beta}^n)$$

be the transition function between boundary charts U_α and U_β , $U_{\alpha\beta} := U_\alpha \cap U_\beta$. Since for any $(x^2, \dots, x^n) \in \mathbb{R}^{n-1}$, $\varphi_{\alpha\beta}^1(0, x^2, \dots, x^n) = 0$.

$$\frac{\partial \varphi_{\alpha\beta}}{\partial x^j} = 0, \quad \text{on } x^1 = 0, \quad j = 2, \dots, n$$

Thus, $\forall p \in \partial M$, the Jacobian has canonical form

$$D(\varphi_{\alpha\beta})_p = \begin{pmatrix} A & 0 \\ * & c \end{pmatrix}$$

where

$$A := \left(\frac{\partial \varphi_{\alpha\beta}^i}{\partial x^j} \right)_{i,j=2,\dots,n}, \quad c = \frac{\partial \varphi_{\alpha\beta}^n}{\partial x^n} > 0$$

Note that this implies $\det D(\varphi_{\alpha\beta})_p = c \det A$, thus, if $D(\varphi_{\alpha\beta})_p > 0$, we have $\det A > 0$, and this is exactly the transition map of the tangent bundle of ∂M . Thus, the change boundary chart preserves orientation. $\square$

Furthermore, if $\omega \in \Omega^n(M)$ is the volume form corresponding to the chosen orientation on M , then the $(n-1)$ -form corresponding to the induced orientation is $\eta := i_{n_{\text{out}}}(\omega|_{\partial M}) \in \Omega^{n-1}(\partial M)$, where n_{out} is the outward pointing vector field. The proof of the fact is simply to consider the local coordinate representation.

Now we may prove Proposition 2.1 that the normal bundle of the boundary is trivial:

Proof of Proposition 2.1. It is enough to construct a smooth outward-pointing vector field along ∂M , since its image in

$$\nu_{\partial M} := TM|_{\partial M} / T\partial M$$

will be a non-vanishing smooth section.

Let $(U_\alpha, \varphi_\alpha)$ be boundary charts with

$$\varphi_\alpha(U_\alpha) \subset \mathbb{H}^n = \{x^1 \geq 0\}, \quad B_\alpha := U_\alpha \cap \partial M.$$

On B_α , define

$$N_\alpha := f_\alpha \frac{\partial}{\partial x_\alpha^1}, \quad f_\alpha \in C^\infty(U_\alpha, \mathbb{R}_{<0}).$$

Then $dx_\alpha^1(N_\alpha) = f_\alpha < 0$, so N_α is outward pointing.

We first note that the outward condition is independent of the chosen boundary chart. On overlaps, write $\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} = (\varphi_{\alpha\beta}^1, \dots, \varphi_{\alpha\beta}^n)$. Since $\varphi_{\alpha\beta}$ preserves the boundary and the interior side,

$$\varphi_{\alpha\beta}^1(0, x^2, \dots, x^n) = 0, \quad x^1 > 0 \rightarrow \varphi_{\alpha\beta}^1(x) > 0.$$

Hence, along the boundary,

$$dx_\alpha^1 = \frac{\partial \varphi_{\alpha\beta}^1}{\partial x_\beta^1} dx_\beta^1, \quad \frac{\partial \varphi_{\alpha\beta}^1}{\partial x_\beta^1} > 0$$

Therefore, $dx_\alpha^1(v) < 0 \iff dx_\beta^1(v) < 0$. Thus, being outward pointing is coordinate-independent.

Now choose a smooth partition of unity $\{\rho_\alpha\}$ on ∂M subordinate to $\{B_\alpha\}$. Define

$$N := \sum_{\alpha} \rho_\alpha N_\alpha.$$

The sum is locally finite, so N is a smooth section of $TM|_{\partial M}$. For any $p \in \partial M$, choose a boundary chart $(U_\gamma, \varphi_\gamma)$ around p . Then

$$dx_\gamma^1(N_p) = \sum_{\alpha} \rho_\alpha(p) dx_\gamma^1(N_{\alpha,p}) < 0,$$

because $\rho_\alpha(p) \geq 0$, $\sum_{\alpha} \rho_\alpha(p) = 1$, and every $N_{\alpha,p}$ with $\rho_\alpha(p) \neq 0$ is outward pointing. Hence N is outward pointing everywhere on ∂M .

In particular, $N_p \neq 0 \forall p \in \partial M$. Moreover, since $T_p \partial M = \ker(dx_\gamma^1)_p$ and $(dx_\gamma^1)_p(N_p) < 0$, we have $N_p \notin T_p \partial M$. Hence the projection $[N] \in \Gamma(\nu_{\partial M})$ is nowhere vanishing. $\square$

With the discussion of the case that the embedded submanifold $N \hookrightarrow M$ is the boundary of M , we may now consider the normal bundle for general submanifolds. The definition of the normal bundle is simply to replace the boundary by a general embedded submanifold in the informal definition given above.

Definition 2.4 (Normal Bundle). Let M be a smooth n -dimensional manifold and let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. We regard TN as a subbundle of i^*TM via di . Then, the normal bundle of N in M is defined by

$$\nu_N(M) := i^*TM/TN$$

Furthermore, a framing of $N \subseteq M$ is a bundle isomorphism

$$\varphi : \nu_N(M) \xrightarrow{\cong} N \times \mathbb{R}^{n-k}$$

i.e., the submanifold N admits a framing if and only if its normal bundle is trivial.

An alternative definition of the normal bundle that shows more aspects of orthogonality is to use a Riemannian metric on M .

Definition 2.5 (Normal Bundle via Riemannian Metric). Let (M, g) be a Riemannian manifold, and let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. The normal bundle on N is given by

$$\nu_N(M, g) := \coprod_{p \in N} (T_p N)^{\perp_g} \subseteq i^*TM = TM|_N$$

where each fiber is the orthogonal complement of $i_{*,p}T_p N$ in $T_{i(p)}M$.

We shall show that these two definitions coincide.

Proposition 2.6. *Let (M, g) be a Riemannian manifold, and let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. Then the natural quotient map*

$$q_g : \nu_N(M, g) \rightarrow \nu_N(M), \quad v \mapsto [v] \mod T_p N$$

where $v \in (T_p N)^{\perp_g}$, is a vector bundle isomorphism.

Proof. For every $p \in N$, since g is positive definite on $T_{i(p)}M$, we have the orthogonal decomposition

$$T_{i(p)}M = i_{*,p}T_pN \oplus (T_pN)^{\perp_g}$$

Hence the induced fiberwise map $q_{g,p} : (T_pN)^{\perp_g} \rightarrow T_{i(p)}M/i_{*,p}T_pN$ is a linear isomorphism.

Indeed, if $v \in (T_pN)^{\perp_g}$ and $q_{g,p}(v) = 0$, then $v \in i_{*,p}T_pN$. Thus

$$v \in i_{*,p}T_pN \cap (T_pN)^{\perp_g}$$

so $g(v, v) = 0$, and therefore $v = 0$. Hence $q_{g,p}$ is injective. On the other hand, for any class $[w] \in T_{i(p)}M/i_{*,p}T_pN$, write $w = u + v$, where $u \in i_{*,p}T_pN$ and $v \in (T_pN)^{\perp_g}$. Then $[w] = [v]$. Hence $q_{g,p}$ is surjective.

It remains to check smoothness. Locally choose a frame $e_1, \dots, e_n$ of i^*TM such that $e_1, \dots, e_k$ is a frame of TN . Let

$$G = (g(e_a, e_b))_{1 \leq a, b \leq k}$$

Since $g|_{TN}$ is positive definite, G is invertible. For $\alpha = k+1, \dots, n$, define

$$\tilde{e}_\alpha = e_\alpha - \sum_{a,b=1}^k (G^{-1})^{ab} g(e_\alpha, e_b) e_a$$

Then, $g(\tilde{e}_\alpha, e_c) = 0$ for every $1 \leq c \leq k$. Hence, $\tilde{e}_{k+1}, \dots, \tilde{e}_n$ is a local frame of $\nu_N(M, g)$. Moreover, $q_g(\tilde{e}_\alpha) = [e_\alpha]$. Since $[e_{k+1}], \dots, [e_n]$ is a local frame of $\nu_N(M)$, the local matrix of q_g is the identity matrix. Therefore, q_g is a smooth vector bundle isomorphism. $\square$

And also, similar to the framing of the boundary, the normal bundle of a general submanifold defines the following short exact sequence:

Proposition 2.7. *The short exact sequence*

$$0 \longrightarrow TN \hookrightarrow i^*TM \twoheadrightarrow \nu_N(M) \longrightarrow 0$$

*admits a metric-dependent splitting. Consequently, $i^*TM \cong TN \oplus \nu_N(M) \cong TN \oplus \nu_N(M, g)$.*

Proof. Let $\pi : i^*TM \rightarrow \nu_N(M)$ be the quotient projection. By the previous proposition,

$$q_g = \pi|_{\nu_N(M, g)} : \nu_N(M, g) \rightarrow \nu_N(M)$$

is a vector bundle isomorphism. Hence

$$s_g := q_g^{-1} : \nu_N(M) \rightarrow \nu_N(M, g) \subseteq i^*TM$$

satisfies $\pi \circ s_g = \text{id}_{\nu_N(M)}$. Thus, s_g is a splitting of the short exact sequence.

More explicitly, define $\Phi_g : TN \oplus \nu_N(M) \rightarrow i^*TM$ by $\Phi_g(u, \eta) = u + s_g(\eta)$. Its inverse is given by

$$\Phi_g^{-1}(w) = (w - s_g(\pi(w)), \pi(w))$$

Indeed,

$$\pi(w - s_g(\pi(w))) = \pi(w) - \pi(s_g(\pi(w))) = \pi(w) - \pi(w) = 0$$

so $w - s_g(\pi(w)) \in \ker \pi = TN$. Therefore, $i^*TM \cong TN \oplus \nu_N(M)$. Since $q_g : \nu_N(M, g) \rightarrow \nu_N(M)$ is an isomorphism, we also obtain $i^*TM \cong TN \oplus \nu_N(M, g)$. $\square$

As a direct corollary of Proposition 2.6, although we use the Riemannian metric to define the normal bundle for technique simplicity, the nature of the normal bundle is topological.

Proposition 2.8. *Let g_0, g_1 be two Riemannian metrics on M . Let*

$$N_0 := \nu_N(M, g_0), \quad N_1 := \nu_N(M, g_1)$$

be the corresponding orthogonal normal bundles. Then there exists a vector bundle isomorphism $N_0 \cong N_1$ that maps the fiber at p to the fiber at p .

The simplest way to prove this proposition is to consider the isomorphism $q_i : \nu_N(M, g_i) \rightarrow \nu_N(M)$. To emphasize the importance of the technique we used in the following proof, we may give an alternative proof using the convex combination of Riemannian metrics.

Proof. Consider the convex combination of metrics $g_t = (1 - t)g_0 + tg_1$ for $t \in [0, 1]$. For every nonzero $v \in T_p M$, we have

$$g_t(v, v) = (1 - t)g_0(v, v) + tg_1(v, v) > 0$$

Hence g_t is a Riemannian metric for every $t \in [0, 1]$.

Let $N_t := \nu_N(M, g_t)$. By the previous proposition, for every $t \in [0, 1]$, the natural quotient map $q_t : N_t \rightarrow \nu_N(M)$ is a vector bundle isomorphism. Therefore we may define $A_t : N_0 \rightarrow N_t$ by $A_t = q_t^{-1} \circ q_0$. This gives a family of vector bundle isomorphisms covering id_N . Moreover, this family varies smoothly in t . Indeed, locally choose a frame $e_1, \dots, e_n$ of i^*TM such that $e_1, \dots, e_k$ is a frame of TN . Let

$$G_t = (g_t(e_a, e_b))_{1 \leq a, b \leq k}.$$

For $\alpha = k + 1, \dots, n$, define

$$\tilde{e}_\alpha(t) = e_\alpha - \sum_{a,b=1}^k (G_t^{-1})^{ab} g_t(e_\alpha, e_b) e_a$$

Then, $\tilde{e}_{k+1}(t), \dots, \tilde{e}_n(t)$ is a local frame of N_t , and it depends smoothly on t . If

$$v = \sum_{\alpha=k+1}^n \lambda_\alpha \tilde{e}_\alpha(0) \in N_0$$

then

$$A_t(v) = \sum_{\alpha=k+1}^n \lambda_\alpha \tilde{e}_\alpha(t)$$

In particular, taking $t = 1$, we get a vector bundle isomorphism $A_1 = q_1^{-1} \circ q_0 : N_0 \rightarrow N_1$. □

And as a corollary of Proposition 2.7, similar to the induced boundary orientation, the

At the end of this section, we may state the most important application of the normal bundle:

Theorem 2.9 (Tubular Neighborhood Theorem). *Let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. Then there exists an open neighborhood $U \subseteq \nu_N(M)$ of the zero section and an open neighborhood $W \subseteq M$ of $i(N)$ such that there is a diffeomorphism $\Phi : U \xrightarrow{\cong} W$ satisfying $\Phi(0_p) = i(p)$ for every $p \in N$.*

Equivalently, after choosing a Riemannian metric g on M , there exists an open neighborhood $U_g \subseteq \nu_N(M, g)$ of the zero section such that the normal exponential map

$$\exp_g^\perp : U_g \rightarrow M, \quad v \mapsto \exp_{i(p)}^g(v), \quad v \in (T_p N)^\perp_g$$

is a diffeomorphism from U_g onto an open neighborhood of $i(N)$ in M .

3 Proof of the Tubular Neighborhood Theorem

In this section, we prove the tubular neighborhood theorem by using the normal exponential map. Let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. For simplicity, we identify N with its image $i(N)$, so that $T_p N \subseteq T_p M$ for every $p \in N$.

Choose a Riemannian metric g on M . Let

$$E := \nu_N(M, g) = \coprod_{p \in N} (T_p N)^\perp_g$$

be the orthogonal normal bundle, and let $\pi : E \rightarrow N$ be the bundle projection.

Definition 3.1 (Normal Exponential Map). Let $\mathcal{D} \subseteq E$ be the set of all $v \in E$ such that the geodesic

$$t \mapsto \exp_{\pi(v)}^g(tv)$$

is defined at $t = 1$. Since geodesics exist locally and depend smoothly on initial conditions, $\mathcal{D}$ is an open neighborhood of the zero section of E .

The normal exponential map is defined by

$$\text{Exp}_g^\perp : \mathcal{D} \rightarrow M, \quad v \mapsto \exp_{\pi(v)}^g(v).$$

The first step is to compute the differential of the normal exponential map along the zero section.

Lemma. *For every $p \in N$, under the canonical identification*

$$T_{0_p} E \cong T_p N \oplus E_p,$$

and under the orthogonal decomposition $T_p M = T_p N \oplus E_p$, we have $d(\text{Exp}_g^\perp)_{0_p}(u, v) = u + v$. In particular, $d(\text{Exp}_g^\perp)_{0_p} : T_{0_p} E \rightarrow T_p M$ is a linear isomorphism.

Proof. Let $u \in T_p N$. Choose a smooth curve $\alpha : (-\epsilon, \epsilon) \rightarrow N$ such that $\alpha(0) = p$ and $\alpha'(0) = u$. Then $\text{Exp}_g^\perp(0_{\alpha(t)}) = \alpha(t)$. Therefore

$$d(\text{Exp}_g^\perp)_{0_p}(u, 0) = \left. \frac{d}{dt} \right|_{t=0} \text{Exp}_g^\perp(0_{\alpha(t)}) = \alpha'(0) = u$$

Now let $v \in E_p = (T_p N)^\perp$. Consider the vertical curve $t \mapsto tv \in E_p$. Then, $\text{Exp}_g^\perp(tv) = \exp_p^g(tv)$. Hence

$$d(\text{Exp}_g^\perp)_{0_p}(0, v) = \frac{d}{dt} \Big|_{t=0} \exp_p^g(tv) = v$$

since the geodesic $t \mapsto \exp_p^g(tv)$ has initial velocity v .

By linearity,

$$d(\text{Exp}_g^\perp)_{0_p}(u, v) = u + v.$$

Since $T_p M = T_p N \oplus E_p$, this map is a linear isomorphism. $\square$

Lemma (Global Inverse Function Lemma Along the Zero Section). *Let $E \rightarrow N$ be a vector bundle, and let $\mathcal{D} \subseteq E$ be an open neighborhood of the zero section. Suppose $F : \mathcal{D} \rightarrow M$ is a smooth map such that $F|_N : N \rightarrow M$ is an embedding and*

$$dF_{0_p} : T_{0_p} E \rightarrow T_{F(0_p)} M$$

is a linear isomorphism for every $p \in N$. Then there exists an open neighborhood $U \subseteq \mathcal{D}$ of the zero section such that

$$F|_U : U \rightarrow F(U)$$

is a diffeomorphism onto an open neighborhood of $F(N)$ in M .

Proof. By the inverse function theorem, for every $p \in N$, there exists an open neighborhood $\Omega_p \subseteq \mathcal{D}$ of 0_p such that

$$F|_{\Omega_p} : \Omega_p \rightarrow F(\Omega_p)$$

is a diffeomorphism onto an open subset of M .

Thus, after replacing $\mathcal{D}$ by a smaller open neighborhood of the zero section, we may assume that F is a local diffeomorphism on $\mathcal{D}$.

It remains to shrink $\mathcal{D}$ so that F becomes injective. Define

$$R = \{(x, y) \in \mathcal{D} \times \mathcal{D} : F(x) = F(y)\}.$$

Since M is Hausdorff, the diagonal $\Delta_M \subseteq M \times M$ is closed, and therefore $R = (F \times F)^{-1}(\Delta_M)$ is closed in $\mathcal{D} \times \mathcal{D}$.

Since F is a local diffeomorphism, for every $x \in \mathcal{D}$ there exists an open neighborhood $\Omega_x \subseteq \mathcal{D}$ such that $F|_{\Omega_x}$ is injective. Hence

$$\Omega := \bigcup_{x \in \mathcal{D}} \Omega_x \times \Omega_x$$

is an open neighborhood of the diagonal $\Delta_{\mathcal{D}}$, and $R \cap \Omega = \Delta_{\mathcal{D}} \cap \Omega$.

Since $F|_N$ is an embedding, it is injective. Hence

$$R \cap (N \times N) = \Delta_N.$$

Therefore, the closed set

$$C := R \cap ((\mathcal{D} \times \mathcal{D}) \setminus \Omega)$$

is disjoint from $N \times N$.

By a standard shrinking argument for manifolds, we may choose an open neighborhood $U \subseteq \mathcal{D}$ of the zero section such that

$$(U \times U) \cap C = \emptyset$$

Now suppose $x, y \in U$ and $F(x) = F(y)$. Then $(x, y) \in R$. Since $(x, y) \notin C$, we must have $(x, y) \in \Omega$. Hence $(x, y) \in \Omega_z \times \Omega_z$ for some $z \in \mathcal{D}$, and $F|_{\Omega_z}$ is injective. Therefore, $x = y$. Thus $F|_U$ is injective.

Since F is a local diffeomorphism on U , the restriction

$$F|_U : U \rightarrow F(U)$$

is a diffeomorphism. Moreover, $F(U)$ is open in M , and since U contains the zero section, $F(U)$ is an open neighborhood of $F(N)$. $\square$

Theorem 3.2 (Tubular Neighborhood Theorem). *Let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. Then there exists an open neighborhood $U \subseteq \nu_N(M)$ of the zero section and an open neighborhood $W \subseteq M$ of $i(N)$ such that there is a diffeomorphism*

$$\Phi : U \xrightarrow{\cong} W$$

satisfying $\Phi(0_p) = i(p)$ for every $p \in N$.

Proof. Choose a Riemannian metric g on M . By the previous discussion, the orthogonal normal bundle $E = \nu_N(M, g)$ is canonically isomorphic to the quotient normal bundle

$$\nu_N(M) = i^*TM/TN$$

via the natural quotient map

$$q_g : E \rightarrow \nu_N(M), \quad v \mapsto [v] \mod T_p N$$

Consider the normal exponential map

$$\text{Exp}_g^\perp : \mathcal{D} \rightarrow M, \quad v \mapsto \exp_{\pi(v)}^g(v)$$

where $\mathcal{D} \subseteq E$ is an open neighborhood of the zero section.

For every $p \in N$, the previous lemma gives

$$d(\text{Exp}_g^\perp)_{0_p}(u, v) = u + v.$$

Since $T_p M = T_p N \oplus E_p$, this differential is a linear isomorphism. Also, $\text{Exp}_g^\perp(0_p) = p$. Hence, the restriction of $\text{Exp}_g^\perp$ to the zero section is precisely the embedding $i : N \hookrightarrow M$.

Therefore, the global inverse function lemma along the zero section applies to $F = \text{Exp}_g^\perp$. Thus, there exists an open neighborhood $U_g \subseteq E = \nu_N(M, g)$ of the zero section such that

$$\text{Exp}_g^\perp|_{U_g} : U_g \rightarrow \text{Exp}_g^\perp(U_g)$$

is a diffeomorphism onto an open neighborhood of N in M .

Let $W := \text{Exp}_g^\perp(U_g)$. Using the bundle isomorphism

$$q_g : \nu_N(M, g) \xrightarrow{\cong} \nu_N(M),$$

set

$$U := q_g(U_g) \subseteq \nu_N(M).$$

Define $\Phi : U \rightarrow W$ by $\Phi(\eta) = \text{Exp}_g^\perp(q_g^{-1}(\eta))$. Then Φ is a diffeomorphism. Moreover, for every $p \in N$,

$$\Phi(0_p) = \text{Exp}_g^\perp(0_p) = i(p).$$

Therefore, U is a tubular neighborhood of the zero section in $\nu_N(M)$, and W is an open neighborhood of $i(N)$ in M diffeomorphic to it. $\square$

Remark 3.3. After choosing a Riemannian metric g on M , the tubular neighborhood diffeomorphism may be written more concretely as

$$\text{Exp}_g^\perp : U_g \subseteq \nu_N(M, g) \xrightarrow{\cong} W \subseteq M.$$

Thus, near N , the manifold M is diffeomorphic to an open neighborhood of the zero section in the normal bundle of N .

4 Regular Values and Preimages

The next bridge between differential topology and homotopy theory is the regular value theorem. It explains how a smooth map produces a submanifold, and in the case of maps to spheres, how a regular preimage naturally carries a framing.

Recall that the regular value of a smooth map $f \in C^\infty(M, N)$ is the value that the preimages admit only the regular points, i.e., the pushforward $(df)_p : T_p M \rightarrow T_p N$ is surjective for all $p \in f^{-1}(q)$.

Definition 4.1 (Regular/Critical Points). For C^∞ -map $f : M \rightarrow N$, $p \in M$ is a regular point if $f_{*,p}$ is surjective. The point $q \in N$ is said to be a regular value if $f^{-1}(q)$ contains only regular points. Otherwise, p is a critical point and $f(p)$ is a critical value.

The most important fact about the regular value that will be repeatedly used in the following passages is that the preimage of a regular value of smooth maps gives an embedded submanifold on the source manifold:

Theorem 4.2 (Preimage of Regular Value). *Let $f : M^m \rightarrow N^n$ with regular value $q \in N$, then*

- $f^{-1}(q)$ is a $(m - n)$ -dimensional embedded submanifold of M .
- $\forall p \in f^{-1}(q)$, the tangent space satisfies $T_p f^{-1}(q) = \ker(df)_p$.

Proof. Consider charts (U, φ) on M and (V, ψ) contains q on N such that $\varphi(p) = \psi(q) = 0$,

$$\tilde{f} := \psi \circ f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}^n, \quad \tilde{f}(0) = 0$$

q is a regular value of $\tilde{f}$ if and only if the Jacobi matrix $J(\tilde{f})$ has rank n at $0 \in \mathbb{R}^m$. With a permutation of coordinates, we shall let $\tilde{f} = (f_1, \dots, f_n)$ with $f_i : \varphi(U) \rightarrow \mathbb{R}^n$

$$\left| \frac{\partial(f_1, \dots, f_n)}{\partial(x^1, \dots, x^n)}(0) \right| \neq 0$$

By inverse function theorem, we shall take $F : \varphi(U) \rightarrow \mathbb{R}^m$ define by

$$F(x^1, \dots, x^m) := (f_1(x), \dots, f_n(x), x^{n+1}, \dots, x^m)$$

and this gives an local diffeomorphism $\varphi(U) \supseteq W \cong W'$ and one shall defined a new chat $(U' := \varphi^{-1}(W), \phi)$ with $\phi = (y^1, \dots, y^m)$ defines by

$$\phi := (F|_W)^{-1} \circ \varphi : U' \rightarrow W', \text{ which explicitly just } \begin{cases} (y^1(x), \dots, y^n(x)) = \tilde{f}(x) \\ (y^{n+1}(x), \dots, y^m(x)) = (x^{n+1}, \dots, x^m) \end{cases}$$

where $f^{-1}(q) \cap U' \cong \tilde{f}^{-1}(0) \cap \psi(U')$ (since $\psi(q) = 0$) has coordinate representation

$$f^{-1}(q) \cap U' := \{p \in U' \mid y^1(p) = \dots = y^n(p) = 0\}$$

By Theorem ??, this shows that $f^{-1}(q)$ is an $(m - n)$ -dimensional submanifold. Furthermore, $f^{-1}(q)$ clearly has tangent space given by

$$T_p f^{-1}(q) := \text{span} \left\{ \frac{\partial}{\partial y^{n+1}} \Big|_p, \dots, \frac{\partial}{\partial y^m} \Big|_p \right\}$$

Since in local coordinate $F(y) = (y^1, \dots, y^n)$,

$$\ker(df)_p \cong \ker(JF) = \{v \in \mathbb{R}^m \mid v^1 = \dots = v^n = 0\}$$

which shows that $\ker(df)_p = \text{span}\{\partial_{y^{n+1}}, \dots, \partial_{y^m}\} = T_p f^{-1}(q)$. □

With the local coordinate representation of pushforward, it is obvious that the rank of the smooth map $f : M \rightarrow N$ at $p \in M$ can be represented by the rank of pushforward $(df)_p : T_p M \rightarrow T_{f(p)} N$ as a linear map. recall that for smooth maps with constant rank

Definition 4.3 (Immersion, Embedding, and Submersion). Let $f : M^m \rightarrow N^n$ be a C^k -map between differentiable manifolds. If $\text{rank}_p f = m \forall p \in M$, f is said to be a C^k -immersion. If $f : M \rightarrow f(M)$ is a C^k -immersion and also a homeomorphism ($f(M) \subseteq N$ has the subset topology of N), then f is a C^k -embedding. If $\text{rank}_p f = n \forall p \in M$, then f is a C^k -submersion.

We may claim the following proposition of differentiable maps with constant rank:

Proposition 4.4. *Let M and N be smooth manifolds. The smooth map $f : M \rightarrow N$ is said to be an immersion if $\forall p \in M : (df)_p : T_p M \rightarrow T_{f(p)} N$ is injective, and a submersion if $\forall p \in M : (df)_p : T_p M \rightarrow T_{f(p)} N$ is surjective (i.e., every points $p \in M$ is a regular point).*

The proof is simply by definition.

A first observation that we can make from Theorem 4.2 is that when $\dim M = \dim N$, the "number of preimages" $|f^{-1}(q)|$ of a regular value is locally constant.

Proposition 4.5. *Let M and N be smooth manifolds with the same dimension, $f : M \rightarrow N$ be a C^∞ -function with regular value $q \in N$, and M compact. Then, there exists an neighborhood $V \subseteq N$ contains q that $\forall q' \in V$*

$$|f^{-1}(q')| = |f^{-1}(q)|$$

Proof. The proof is simply by the inverse function theorem. By the inverse function theorem, $\forall p_i \in f^{-1}(q)$, there exists an open neighborhood U_i such that $f|_{U_i} : U_i \rightarrow V_i \subseteq N$ is a diffeomorphism. Then, we need the following lemma:

Lemma. *Let M and N be smooth manifolds with the same dimension. For M compact and $\forall f \in C^\infty(M, N)$, if $q \in N$ is a regular value, then*

$$f^{-1}(q) = \{p_1, \dots, p_k\}$$

is a finite subset in M .

The proof of the lemma simply follows the fact that since $q \in N$ is a regular value, $\dim f^{-1}(q) = 0$, i.e., $f^{-1}(q)$ is discrete and closed (since the singleton $\{q\}$ is a closed set in N). By the compactness of M , since discrete subset $f^{-1}(q)$ is closed, $f^{-1}(q)$ is compact, and thus, is a finite set.

By the Hausdorffness of manifolds, one shall always let $U_i, \dots, U_k$ be disjoint and diffeomorphic to $V_1, \dots, V_k$. Then, in the open set

$$V = \left(\bigcap_{i=1}^k V_i \right) \setminus f\left(M \setminus \bigcup_{i=1}^k U_i\right)$$

Every point $q' \in V$ has number of preimage $|f^{-1}(q')| = k$. □

The set of critical values, in the case that $\dim M = \dim N$, is also finite.

Proposition 4.6. *Let $f : M \rightarrow N$ be a smooth map between smooth manifolds. Suppose that M is compact and $\dim M = \dim N$. If $y \in N$ is a regular value of f , then $f^{-1}(y)$ is a finite set.*

Proof. Since y is a regular value of f , the regular value theorem implies that $f^{-1}(y)$ is a smooth submanifold of M of dimension

$$\dim f^{-1}(y) = \dim M - \dim N.$$

Because $\dim M = \dim N$, we have $\dim f^{-1}(y) = 0$. Thus $f^{-1}(y)$ is a 0-dimensional smooth manifold, hence a discrete topological space.

On the other hand, since f is continuous and $\{y\} \subset N$ is closed, the set $f^{-1}(y)$ is closed in M . Since M is compact, it follows that $f^{-1}(y)$ is compact.

Therefore, $f^{-1}(y)$ is both compact and discrete. A compact discrete space must be finite. Indeed, if $f^{-1}(y)$ were infinite, then the open cover

$$\{\{x\} : x \in f^{-1}(y)\}$$

would have no finite subcover, contradicting compactness. Hence $f^{-1}(y)$ is finite. □

In general, when $\dim M \neq \dim N$, one should not expect the preimage of a regular value to be a nonempty finite set. But we can still expect it to be "small" in the sense of measure.

Theorem 4.7 (Sard's Theorem). *For any $f \in C^\infty(M, N)$, the set of critical points of f has measure-zero in N .*

The proof of Sard's theorem can be seen in [1], and we will not include the proof here.

Finally, follows from Theorem 4.1, note that for $f : M \rightarrow N$ and $p \in f^{-1}(q)$ the following sequence

$$0 \longrightarrow T_p f^{-1}(q) \hookrightarrow T_p M \xrightarrow{(df)_p} T_q N \longrightarrow 0$$

is a short exact sequence. Thus, the sequence above leads to the framing of the preimage $f^{-1}(q)$ since the image $q \in N$ is a fixed point, i.e., $\forall p \in f^{-1}(q), \nu_{f^{-1}(q)}(M)_p \cong T_q N$, and the fiberwise isomorphism can be safely combined into the bundle isomorphism. But that is just

$$\nu_{f^{-1}(q)}(M) \cong f^{-1}(q) \times T_q N$$

which leads to the framing on $f^{-1}(q)$ in M .

5 Framed Cobordism

The previous section shows that the preimage of a regular value is not merely an embedded submanifold. More importantly, it has a canonical normal bundle description. In particular, if $f : M^m \rightarrow N^r$ is a smooth map and $q \in N$ is a regular value, then

$$\nu_{f^{-1}(q)}(M) \cong f^{-1}(q) \times T_q N.$$

Thus, after choosing a basis of the fixed vector space $T_q N$, the regular preimage $f^{-1}(q)$ becomes a framed submanifold of M . The purpose of this section is to define the equivalence relation on such framed submanifolds that corresponds to the homotopy of maps.

Throughout this section, M^m is a smooth manifold without boundary unless otherwise stated.

Definition 5.1 (Framed Submanifold). Let $X^{m-r} \subseteq M^m$ be a compact embedded submanifold of codimension r . The normal bundle of X in M is defined by

$$\nu_X(M) := TM|_X / TX.$$

A framing of X in M is a vector bundle isomorphism $\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$. A framed submanifold of M is a pair (X, φ) , where $X \subseteq M$ is a compact embedded submanifold and φ is a framing of its normal bundle.

Equivalently, after choosing a Riemannian metric on M and identifying $\nu_X(M) \cong TX^\perp$, a framing may be represented by r smooth normal vector fields $(v_1, \dots, v_r) \in \Gamma(TX^\perp)^{\oplus r}$ such that for every $x \in X$, the ordered tuple $(v_1(x), \dots, v_r(x))$ is a basis of the normal space $\nu_X(M)_x$. Hence, a framing is stronger than an orientation of the normal bundle: it trivializes the whole normal bundle, not just its determinant line.

We also allow the empty submanifold $\emptyset \subseteq M$. It has a unique framing and will be regarded as a framed submanifold.

Proposition 5.2. *Let M be an oriented smooth manifold and let (X, φ) be a framed submanifold of M . Then the framing φ induces an orientation on X .*

Proof. Since $X \subseteq M$ is an embedded submanifold, we have a short exact sequence of vector bundles over X :

$$0 \longrightarrow TX \hookrightarrow TM|_X \longrightarrow \nu_X(M) \longrightarrow 0.$$

Taking determinant lines gives the canonical isomorphism

$$\det(TM|_X) \cong \det(TX) \otimes \det(\nu_X(M)).$$

The orientation of M gives an orientation of $\det(TM|_X)$, while the framing

$$\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

gives an orientation of $\nu_X(M)$ by pulling back the standard orientation of $\mathbb{R}^r$. Therefore, the determinant-line isomorphism above uniquely determines an orientation of $\det(TX)$, and hence an orientation of X . $\square$

The main equivalence relation is the following.

Definition 5.3 (Framed Cobordism). Let (X_0, φ_0) and (X_1, φ_1) be compact framed submanifolds of M^m of codimension r . We say that (X_0, φ_0) and (X_1, φ_1) are framed cobordant if there exists a compact, properly embedded submanifold with boundary

$$W^{m-r+1} \subseteq M \times [0, 1]$$

such that $\partial W = X_0 \times \{0\} \sqcup X_1 \times \{1\}$ as a set. If orientations are included, this identity is understood as

$$\partial W = (-X_0) \times \{0\} \sqcup X_1 \times \{1\}$$

Moreover, W is required to be cylindrical near the boundary, i.e., there exists $\varepsilon > 0$ such that

$$W \cap (M \times [0, \varepsilon)) = X_0 \times [0, \varepsilon) \quad \text{and} \quad W \cap (M \times (1 - \varepsilon, 1]) = X_1 \times (1 - \varepsilon, 1]$$

Finally, W is equipped with a framing $\Phi : \nu_W(M \times [0, 1]) \xrightarrow{\cong} W \times \mathbb{R}^r$ whose restrictions to the boundary agree with φ_0 and φ_1 .

Let us explain the last condition more explicitly. Near the boundary, W is a product. Hence, along $X_i \times \{i\}$, $i = 0, 1$, we have

$$TW|_{X_i \times \{i\}} = TX_i \oplus \mathbb{R}\partial_t$$

while

$$T(M \times [0, 1])|_{X_i \times \{i\}} = TM|_{X_i} \oplus \mathbb{R}\partial_t$$

Therefore,

$$\begin{aligned} \nu_W(M \times [0, 1])|_{X_i \times \{i\}} &\cong \frac{TM|_{X_i} \oplus \mathbb{R}\partial_t}{TX_i \oplus \mathbb{R}\partial_t} \cong \frac{TM|_{X_i}}{TX_i} \\ &= \nu_{X_i}(M) \end{aligned}$$

Under this canonical identification, the restriction of Φ to $X_i \times \{i\}$ is required to be φ_i . In other words,

$$\Phi|_{X_0 \times \{0\}} = \varphi_0, \quad \Phi|_{X_1 \times \{1\}} = \varphi_1$$

If such a framed cobordism exists, we write $(X_0, \varphi_0) \sim (X_1, \varphi_1)$.

Proposition 5.4. *Framed cobordism is an equivalence relation.*

Proof. We verify reflexivity, symmetry, and transitivity.

First, reflexivity follows from the product cobordism. For a framed submanifold (X, φ) , take $W := X \times [0, 1] \subseteq M \times [0, 1]$. Then,

$$\partial W = X \times \{0\} \sqcup X \times \{1\}$$

and W is cylindrical near the boundary. Moreover,

$$\nu_{X \times [0, 1]}(M \times [0, 1]) \cong \pi_X^* \nu_X(M),$$

where $\pi_X : X \times [0, 1] \rightarrow X$ is the projection. Hence, the framing φ pulls back to a framing $\Phi := \pi_X^* \varphi$ of W . Therefore, $(X, \varphi) \sim (X, \varphi)$.

Second, suppose $W \subseteq M \times [0, 1]$ is a framed cobordism from (X_0, φ_0) to (X_1, φ_1) . Consider the diffeomorphism

$$R : M \times [0, 1] \rightarrow M \times [0, 1], \quad R(x, t) = (x, 1 - t).$$

Then $R(W)$ is a cobordism from X_1 to X_0 . The framing of W is carried forward by R to a framing of $R(W)$. Thus, $(X_1, \varphi_1) \sim (X_0, \varphi_0)$.

Finally, suppose

$$(X_0, \varphi_0) \sim (X_1, \varphi_1), \quad (X_1, \varphi_1) \sim (X_2, \varphi_2)$$

Let W_{01} be a framed cobordism from (X_0, φ_0) to (X_1, φ_1) and let W_{12} be a framed cobordism from (X_1, φ_1) to (X_2, φ_2) . After rescaling the interval, we may regard W_{01} as lying in $M \times [0, \frac{1}{2}]$ and W_{12} as lying in $M \times [\frac{1}{2}, 1]$. Since both cobordisms are cylindrical near X_1 , the union

$$W_{02} := W_{01} \cup_{X_1} W_{12}$$

is a smooth properly embedded submanifold of $M \times [0, 1]$. The framings also glue smoothly because both of them restrict to the same framing φ_1 on X_1 . Therefore, W_{02} is a framed cobordism from (X_0, φ_0) to (X_2, φ_2) . Hence, $(X_0, \varphi_0) \sim (X_2, \varphi_2)$. This proves that framed cobordism is an equivalence relation. $\square$

Definition 5.5 (Framed Cobordism Classes). The set of framed cobordism classes of compact codimension r framed submanifolds of M^m is denoted by

$$\Omega_{m-r}^{\text{fr}}(M)$$

Here, the subscript records the dimension of the framed submanifold. The codimension r is kept implicit in the notation.

At this point, $\Omega_{m-r}^{\text{fr}}(M)$ should be regarded as a set of equivalence classes. In special situations, such as $M = S^m$, this set will later be identified with a homotopy set

$$[S^m, S^r] = \pi_m(S^r)$$

which carries the usual homotopy group structure when $m \geq 1$ and $r \geq 1$.

We now explain how framed submanifolds arise naturally from regular preimages. The following proposition has been informally stated at the end of the last section:

Proposition 5.6 (Regular Preimages are Framed). *Let $f : M^m \rightarrow N^r$ be a smooth map and let $q \in N$ be a regular value of f . Then $X := f^{-1}(q)$ is a codimension r embedded submanifold of M . Moreover, after choosing an ordered basis $\beta = (e_1, \dots, e_r)$ of $T_q N$, the submanifold X has a canonical framing*

$$\varphi_{f,q,\beta} : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

Proof. By Theorem 4.2, since q is a regular value, $X = f^{-1}(q)$ is a smooth embedded submanifold of dimension $m - r$, and $T_p X = \ker(df)_p$ for every $p \in X$. Hence, we have a short exact sequence

$$0 \longrightarrow T_p X \hookrightarrow T_p M \xrightarrow{(df)_p} T_q N \longrightarrow 0$$

Taking the quotient by $T_p X$, the differential induces a linear isomorphism

$$\overline{(df)_p} : T_p M / T_p X \xrightarrow{\cong} T_q N$$

Since $T_p M / T_p X = \nu_X(M)_p$, this gives $\nu_X(M)_p \cong T_q N$ for every $p \in X$. Since q is fixed, these fiberwise isomorphisms assemble into a smooth vector bundle isomorphism $\nu_X(M) \cong X \times T_q N$.

Finally, the chosen basis $\beta = (e_1, \dots, e_r)$ identifies $T_q N \cong \mathbb{R}^r$. Therefore,

$$\nu_X(M) \cong X \times T_q N \cong X \times \mathbb{R}^r$$

which is the required framing. □

Explicitly, the framing in Proposition 5.6 is given by

$$\varphi_{f,q,\beta}(p, [v]) = (p, [(df)_p(v)]_\beta),$$

where $[v] \in T_p M / T_p X$ and $[(df)_p(v)]_\beta$ denotes the coordinate vector of $(df)_p(v) \in T_q N$ with respect to the basis β .

For the Pontryagin-Thom construction, the most important case is $f : M^m \rightarrow S^r$. If $q \in S^r$ is a regular value, then the framed submanifold is $f^{-1}(q) \subseteq M$, with normal framing induced by the differential

$$(df)_p : T_p M \rightarrow T_q S^r.$$

The next proposition explains why framed cobordism is the correct equivalence relation for regular preimages.

Theorem 5.7 (Relative Transversality). *Let A be a smooth manifold with boundary and let $g : A \rightarrow N$ be a smooth map. Suppose that g is already transverse to a submanifold $Y \subseteq N$ on a neighborhood of ∂A . Then g can be perturbed, while fixed on a neighborhood of ∂A , to a smooth map $\tilde{g} : A \rightarrow N$ such that $\tilde{g} \pitchfork Y$.*

We will use this theorem without proof. It is a standard relative version of the transversality theorem.

Proposition 5.8 (Homotopy Gives Framed Cobordism). *Let M^m be compact. Let*

$$f_0, f_1 : M \rightarrow S^r$$

be smooth maps, and let $q \in S^r$ be a regular value of both f_0 and f_1 . Suppose that f_0 and f_1 are smoothly homotopic. Then, after choosing a basis of $T_q S^r$, the framed submanifolds $f_0^{-1}(q)$, $f_1^{-1}(q)$ are framed cobordant.

Proof. Let $H : M \times [0, 1] \rightarrow S^r$ be a smooth homotopy from f_0 to f_1 , so that

$$H(-, 0) = f_0, \quad H(-, 1) = f_1$$

First, choose a smooth function $\rho : [0, 1] \rightarrow [0, 1]$ such that

$$\rho(t) = 0 \quad \text{for } t \in [0, \varepsilon), \quad \rho(t) = 1 \quad \text{for } t \in (1 - \varepsilon, 1]$$

for some sufficiently small $\varepsilon > 0$. Replacing H by

$$H'(x, t) := H(x, \rho(t))$$

We may assume that the homotopy is constant near $t = 0$ and $t = 1$.

Since q is a regular value of both f_0 and f_1 , the map H' is already transverse to q near the boundary of $M \times [0, 1]$. By Theorem 5.7, after a small perturbation fixed near the boundary, we may assume that $H' \pitchfork \{q\}$. For simplicity, we still denote the perturbed homotopy by H .

Define $W := H^{-1}(q) \subseteq M \times [0, 1]$. Since H is transverse to q , the regular value theorem implies that W is a smooth embedded submanifold of codimension r . Since M is compact, W is compact. Moreover, because H is constant near $t = 0$ and $t = 1$, we have

$$W \cap (M \times [0, \varepsilon)) = f_0^{-1}(q) \times [0, \varepsilon) \quad \text{and} \quad W \cap (M \times (1 - \varepsilon, 1]) = f_1^{-1}(q) \times (1 - \varepsilon, 1]$$

Thus,

$$\partial W = f_0^{-1}(q) \times \{0\} \sqcup f_1^{-1}(q) \times \{1\}$$

It remains to construct the framing on W . For every $z = (x, t) \in W$, transversality gives a short exact sequence

$$0 \longrightarrow T_z W \hookrightarrow T_z(M \times [0, 1]) \xrightarrow{(dH)_z} T_q S^r \longrightarrow 0.$$

Hence, $\nu_W(M \times [0, 1])_z \cong T_q S^r$. Since q is fixed, these fiberwise isomorphisms assemble into a vector bundle isomorphism

$$\nu_W(M \times [0, 1]) \cong W \times T_q S^r$$

Choosing a basis of $T_q S^r$ gives a framing $\Phi : \nu_W(M \times [0, 1]) \xrightarrow{\cong} W \times \mathbb{R}^r$.

On the boundary component $f_0^{-1}(q) \times \{0\}$, the differential (dH) agrees with (df_0) in the M -direction. Therefore, the induced framing restricts to the regular preimage framing of $f_0^{-1}(q)$. Similarly, on the boundary component $f_1^{-1}(q) \times \{1\}$, it restricts to the regular preimage framing of $f_1^{-1}(q)$. Therefore, W is a framed cobordism from $f_0^{-1}(q)$ to $f_1^{-1}(q)$. $\square$

Consequently, the framed cobordism class of a regular preimage depends only on the homotopy class of the original map. More precisely, if $f_0 \simeq f_1 : M \rightarrow S^r$, then

$$[f_0^{-1}(q)]_{\text{fr}} = [f_1^{-1}(q)]_{\text{fr}} \in \Omega_{m-r}^{\text{fr}}(M)$$

provided q is chosen as a regular value of the relevant maps after a possible small perturbation.

The converse direction, namely the construction of a map from a framed submanifold, will be the content of the Pontryagin-Thom construction. The key geometric input is the tubular neighborhood theorem.

Proposition 5.9 (Framed Tubular Neighborhood). *Let (X, φ) be a codimension r framed submanifold of M^m . Then there exists a tubular neighborhood U of X in M and a diffeomorphism*

$$\Theta : X \times D^r \xrightarrow{\cong} U$$

such that $\Theta(x, 0) = x$ for every $x \in X$. Moreover, the derivative of Θ in the D^r -direction realizes the chosen framing φ .

Proof. By the tubular neighborhood theorem, there exists an open neighborhood U of X in M and a diffeomorphism from a neighborhood of the zero section in $\nu_X(M)$ onto U . Since X is framed, we have a vector bundle isomorphism

$$\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

Shrinking the tubular neighborhood if necessary, we may restrict to a disk bundle in $\nu_X(M)$ which corresponds under φ to $X \times D^r \subseteq X \times \mathbb{R}^r$. Composing the inverse of this trivialization with the tubular neighborhood diffeomorphism gives

$$\Theta : X \times D^r \xrightarrow{\cong} U$$

By construction, $\Theta(x, 0) = x$, and the normal coordinate directions in D^r agree with the chosen framing. $\square$

Similarly, if $(W, \Phi) \subseteq M \times [0, 1]$ is a framed cobordism, then the tubular neighborhood theorem and the framing Φ give a product tubular neighborhood $W \times D^r \subseteq M \times [0, 1]$.

This compatibility is the geometric reason that framed cobordism corresponds to homotopy. A framed submanifold produces a map by collapsing the complement of a framed tubular neighborhood, and a framed cobordism produces a homotopy between the corresponding collapse maps.

The next section will turn this geometric relation into the Pontryagin-Thom correspondence.

6 The Pontryagin-Thom Construction

The previous section introduced framed cobordism as the geometric equivalence relation on framed submanifolds. We now explain why this equivalence relation is naturally related to homotopy classes of maps to spheres.

The main idea is simple. If (X, φ) is a codimension r framed submanifold of M , then a framed tubular neighborhood identifies a neighborhood of X with $X \times D^r$. The Pontryagin-Thom construction sends a point in this tubular neighborhood to its normal coordinate in D^r , and sends every point outside the tubular neighborhood to the base point of

$$S^r \cong D^r / \partial D^r$$

Thus, a framed submanifold produces a map $M \rightarrow S^r$.

Conversely, if $f : M \rightarrow S^r$ is a smooth map and $q \in S^r$ is a regular value, then $f^{-1}(q)$ is a framed submanifold of M . These two constructions are inverse to each other up to framed cobordism and homotopy.

Throughout this section, let M^m be a compact smooth manifold without boundary. We write

$$S^r := D^r / \partial D^r$$

and denote the image of ∂D^r by $*$, the base point of S^r .

Definition 6.1 (Pontryagin-Thom Collapse Map). Let (X, φ) be a compact codimension r framed submanifold of M^m . Choose a framed tubular neighborhood

$$\Theta : X \times D^r \xrightarrow{\cong} U \subseteq M$$

such that $\Theta(x, 0) = x$. The Pontryagin-Thom collapse map associated to (X, φ) is the continuous map

$$c_{X, \varphi} : M \rightarrow S^r = D^r / \partial D^r$$

defined by

$$c_{X, \varphi}(p) = \begin{cases} [v], & p = \Theta(x, v) \in X \times D^r, \\ *, & p \in M \setminus \mathring{U}. \end{cases}$$

Here $[v]$ denotes the image of $v \in D^r$ under the quotient map $D^r \rightarrow D^r / \partial D^r$. The definition is continuous because if p approaches the boundary of the tubular neighborhood, then the normal coordinate v approaches ∂D^r , whose image in the quotient is exactly the base point $*$.

In particular,

$$c_{X, \varphi}^{-1}(0) = X,$$

where $0 \in D^r \subseteq D^r / \partial D^r$ denotes the image of the origin. Thus, the framed submanifold X is recovered as the inverse image of the center of the disk.

Proposition 6.2. *Let (X, φ) be a codimension r framed submanifold of M . Then $0 \in D^r / \partial D^r$ is a regular value of the collapse map $c_{X, \varphi}$ after smoothing the collapse map near ∂U , and the regular preimage framing of*

$$c_{X, \varphi}^{-1}(0) = X$$

agrees with the original framing φ .

Proof. On the tubular neighborhood $U \cong X \times D^r$, the collapse map is given by projection to the normal coordinate:

$$c_{X, \varphi}(x, v) = [v].$$

Near the zero section, this is simply the projection $X \times D^r \rightarrow D^r$. Therefore, $c_{X, \varphi}^{-1}(0) = X \times \{0\} \cong X$.

At a point $(x, 0) \in X \times D^r$, the differential of the local projection is

$$T_x X \oplus T_0 D^r \rightarrow T_0 D^r, \quad (\xi, \eta) \mapsto \eta.$$

Hence, it is surjective, and its kernel is exactly $T_x X$. Therefore, 0 is a regular value near X .

The induced normal bundle isomorphism is $\nu_X(M) \cong X \times T_0 D^r$. Under the identification $T_0 D^r \cong \mathbb{R}^r$, this is precisely the framing determined by the tubular neighborhood and the chosen trivialization

$$\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

Thus, the regular preimage framing agrees with φ .

The possible non-smoothness of $c_{X,\varphi}$ occurs only near ∂U , where the quotient collapses ∂D^r to the base point. One may replace the radial coordinate by a smooth bump function which is constant near ∂D^r . This smoothing does not affect the map near $X \times \{0\}$, and hence does not change the regular preimage or its framing. $\square$

We next show that the homotopy class of the collapse map does not depend on the auxiliary choices.

Proposition 6.3. *The homotopy class of the Pontryagin-Thom collapse map*

$$[c_{X,\varphi}] \in [M, S^r]$$

depends only on the framed submanifold (X, φ) , not on the chosen framed tubular neighborhood.

Proof. Suppose $\Theta_0 : X \times D^r \rightarrow U_0$ and $\Theta_1 : X \times D^r \rightarrow U_1$ are two framed tubular neighborhoods compatible with the same framing φ . Since both tubular neighborhoods realize the same normal bundle trivialization along the zero section, they are isotopic after possibly shrinking the disk radius. Thus, there exists a smooth family of framed tubular neighborhoods

$$\Theta_t : X \times D^r \rightarrow U_t, \quad t \in [0, 1],$$

with Θ_0 and Θ_1 as endpoints, after replacing D^r by a smaller disk if necessary.

Applying Definition 6.1 to Θ_t gives a continuous family of maps

$$c_t : M \rightarrow S^r.$$

This family is a homotopy from $c_{X,\varphi}^{(0)}$ to $c_{X,\varphi}^{(1)}$. Hence the homotopy class of the collapse map is independent of the chosen framed tubular neighborhood. $\square$

Therefore, a framed submanifold determines a well-defined homotopy class of maps to S^r .

Definition 6.4 (The Pontryagin-Thom Map). Define

$$\text{PT} : \Omega_{m-r}^{\text{fr}}(M) \rightarrow [M, S^r]$$

by $\text{PT}([X, \varphi]) := [c_{X,\varphi}]$, where $c_{X,\varphi} : M \rightarrow S^r$ is the Pontryagin-Thom collapse map.

We must check that this is well-defined on framed cobordism classes.

Proposition 6.5. *If two framed submanifolds (X_0, φ_0) and (X_1, φ_1) are framed cobordant in M , then their Pontryagin-Thom collapse maps are homotopic, i.e., $c_{X_0, \varphi_0} \simeq c_{X_1, \varphi_1}$.*

Proof. Let $W^{m-r+1} \subseteq M \times [0, 1]$ be a framed cobordism from (X_0, φ_0) to (X_1, φ_1) , with framing

$$\Phi : \nu_W(M \times [0, 1]) \xrightarrow{\cong} W \times \mathbb{R}^r$$

By the framed tubular neighborhood theorem, there is a tubular neighborhood

$$\Theta : W \times D^r \xrightarrow{\cong} U_W \subseteq M \times [0, 1]$$

compatible with the framing Φ .

Since W is cylindrical near the boundary, the tubular neighborhood can also be chosen cylindrical near the boundary. That is, for sufficiently small $\varepsilon > 0$, near $t = 0$ it agrees with $X_0 \times [0, \varepsilon) \times D^r$, and near $t = 1$ it agrees with

$$X_1 \times (1 - \varepsilon, 1] \times D^r.$$

Define a collapse map $C : M \times [0, 1] \rightarrow S^r$ by

$$C(p, t) = \begin{cases} [v], & (p, t) = \Theta(w, v) \in W \times D^r, \\ *, & (p, t) \notin \mathring{U}_W \end{cases}$$

This is continuous for the same reason as in Definition 6.1. Since the tubular neighborhood is cylindrical near the boundary, the restriction of C to $M \times \{0\}$ is the collapse map associated to (X_0, φ_0) , and the restriction of C to $M \times \{1\}$ is the collapse map associated to (X_1, φ_1) . Therefore, C is a homotopy $c_{X_0, \varphi_0} \simeq c_{X_1, \varphi_1}$. $\square$

Hence, the map $\text{PT} : \Omega_{m-r}^{\text{fr}}(M) \rightarrow [M, S^r]$ is well-defined.

Now we construct the inverse direction. Let $f : M \rightarrow S^r$ be a smooth map. By Sard's theorem, regular values exist densely in S^r . Choose a regular value $q \in S^r \setminus \{*\}$. Then $f^{-1}(q)$ is a codimension r submanifold of M . Since q is fixed, the differential of f gives a canonical normal bundle identification

$$\nu_{f^{-1}(q)}(M) \cong f^{-1}(q) \times T_q S^r.$$

After choosing a basis of $T_q S^r$, this becomes a framing.

Definition 6.6 (Regular Preimage Construction). Let $f : M \rightarrow S^r$ be a smooth map and let $q \in S^r \setminus \{*\}$ be a regular value. Choose an ordered basis β of $T_q S^r$. Define

$$\text{RP}(f) := (f^{-1}(q), \varphi_{f, q, \beta}),$$

where $\varphi_{f, q, \beta}$ is the framing induced by $(df)_p : T_p M \rightarrow T_q S^r$ for $p \in f^{-1}(q)$.

The notation RP stands for regular preimage.

Proposition 6.7. *The framed cobordism class of $\text{RP}(f)$ depends only on the homotopy class of f .*

Proof. Suppose $f_0 \simeq f_1 : M \rightarrow S^r$. Choose a smooth homotopy $H : M \times [0, 1] \rightarrow S^r$ from f_0 to f_1 . After a perturbation relative to the boundary, we may assume that H is transverse to the chosen regular value q . Then, $W := H^{-1}(q)$ is a compact embedded submanifold of $M \times [0, 1]$ with boundary $\partial W = f_0^{-1}(q) \times \{0\} \sqcup f_1^{-1}(q) \times \{1\}$.

Moreover, for every $z \in W$, transversality gives a short exact sequence

$$0 \longrightarrow T_z W \hookrightarrow T_z(M \times [0, 1]) \xrightarrow{(dH)_z} T_q S^r \longrightarrow 0$$

Hence, $\nu_W(M \times [0, 1]) \cong W \times T_q S^r$. Choosing the same basis of $T_q S^r$ gives a framing of W . Its boundary restrictions agree with the regular preimage framings of $f_0^{-1}(q)$ and $f_1^{-1}(q)$. Therefore, $\text{RP}(f_0) \sim \text{RP}(f_1)$. $\square$

The construction is also independent of the chosen regular value, up to framed cobordism.

Proposition 6.8. *Let $q_0, q_1 \in S^r \setminus \{*\}$ be regular values of $f : M \rightarrow S^r$. After choosing compatible bases of $T_{q_0}S^r$ and $T_{q_1}S^r$, the framed submanifolds $f^{-1}(q_0)$ and $f^{-1}(q_1)$ are framed cobordant.*

Proof. Choose a diffeomorphism $a : S^r \rightarrow S^r$ isotopic to the identity such that $a(q_0) = q_1$. Then $a \circ f \simeq f$.

Since q_1 is a regular value of $a \circ f$ precisely when q_0 is a regular value of f , we have

$$(a \circ f)^{-1}(q_1) = f^{-1}(q_0).$$

By Proposition 6.7, the regular preimage of $a \circ f$ at q_1 is framed cobordant to the regular preimage of f at q_1 . Therefore, $f^{-1}(q_0) \sim f^{-1}(q_1)$. $\square$

We now state the main correspondence.

Theorem 6.9 (Pontryagin-Thom Correspondence). *Let M^m be a compact smooth manifold without boundary. Then the Pontryagin-Thom construction gives a natural bijection*

$$\text{PT} : \Omega_{m-r}^{\text{fr}}(M) \xrightarrow{\cong} [M, S^r].$$

Proof. We construct inverse maps.

First, starting from a framed submanifold (X, φ) , Definition 6.1 gives the collapse map $c_{X, \varphi} : M \rightarrow S^r$. Thus, we have a map

$$\text{PT} : \Omega_{m-r}^{\text{fr}}(M) \rightarrow [M, S^r]$$

Conversely, starting from a homotopy class $[f] \in [M, S^r]$, choose a smooth representative $f : M \rightarrow S^r$ and a regular value $q \in S^r \setminus \{*\}$. The regular preimage construction gives a framed submanifold

$$\text{RP}(f) = (f^{-1}(q), \varphi_{f,q})$$

By Propositions 6.7 and 6.8, its framed cobordism class depends only on $[f]$. Hence, we obtain a well-defined map

$$\text{RP} : [M, S^r] \rightarrow \Omega_{m-r}^{\text{fr}}(M).$$

It remains to prove that PT and RP are inverse to each other.

First, let (X, φ) be a framed submanifold. Consider its collapse map $c_{X, \varphi} : M \rightarrow S^r$. By Proposition 6.2, $c_{X, \varphi}^{-1}(0) = X$, and the regular preimage framing agrees with the original framing φ . Therefore,

$$\text{RP}(\text{PT}([X, \varphi])) = [X, \varphi]$$

Second, let $f : M \rightarrow S^r$ be a smooth map, and let $q \in S^r \setminus \{*\}$ be a regular value. We need to compare f with the collapse map associated with the framed submanifold $X = f^{-1}(q)$.

Choose a small coordinate ball $B \subseteq S^r$ centered at q , and choose coordinates $B \cong D^r$ sending q to 0. Since q is a regular value, the map f is a submersion near X . Hence, after shrinking B , the preimage $f^{-1}(B)$ is a tubular neighborhood of X , and the map f on this neighborhood is identified with the normal coordinate projection $X \times D^r \rightarrow D^r$.

Now choose a map $h : S^r \rightarrow S^r$ which is homotopic to the identity, sends $S^r \setminus B$ to the base point $*$, and agrees with the quotient projection

$$D^r \rightarrow D^r / \partial D^r$$

on $B \cong D^r$. Such a map has degree one, and hence is homotopic to the identity.

Then, $h \circ f \simeq f$. But $h \circ f$ is exactly the Pontryagin-Thom collapse map associated to the framed submanifold $X = f^{-1}(q)$. Therefore

$$\text{PT}(\text{RP}([f])) = [f]$$

Hence PT and RP are inverse bijections. □

Let us emphasize the two directions of the correspondence:

$$\begin{aligned} (X, \varphi) \subseteq M &\longmapsto c_{X, \varphi} : M \rightarrow S^r \\ f : M \rightarrow S^r &\longmapsto f^{-1}(q) \subseteq M \end{aligned}$$

where q is a regular value of f .

In the special case $M = S^m$, the correspondence becomes

$$\Omega_{m-r}^{\text{fr}}(S^m) \cong [S^m, S^r] = \pi_m(S^r)$$

Thus, homotopy groups of spheres can be interpreted geometrically as framed cobordism classes of submanifolds of spheres.

This is the fundamental bridge between differential topology and unstable homotopy theory:

$$\text{homotopy classes of maps to spheres} \iff \text{framed cobordism classes}$$

The next section applies this viewpoint to concrete examples, including the computation of low-dimensional stable stems and the relation with degree, vector fields, and the Poincaré-Hopf theorem.

7 Applications and Examples

7.1 The Freudenthal Suspension Theorem

In this subsection, we prove the sphere version of the Freudenthal suspension theorem by using the Pontryagin-Thom correspondence. The geometric idea is that the suspension homomorphism is represented by adding one trivial normal direction to a framed submanifold. The stable range then comes from the fact that, under suitable dimension bounds, a framed submanifold in $\mathbb{R}^m \oplus \mathbb{R}$ may be pushed into the hyperplane $\mathbb{R}^m$ and the extra normal direction may be split off from the framing.

For integers $n \geq 1$ and $k \geq 0$, let $\mathcal{F}_k^n$ denote the set of framed cobordism classes of compact codimension n framed submanifolds of $\mathbb{R}^{n+k}$.

Proposition 7.1. *There is a natural bijection*

$$\text{PT} : \mathcal{F}_k^n \xrightarrow{\cong} \pi_{n+k}(S^n).$$

Proof. Identify S^{n+k} with the one-point compactification

$$(\mathbb{R}^{n+k})^+ = \mathbb{R}^{n+k} \sqcup \{\infty\}$$

By Theorem 6.9, framed cobordism classes of compact codimension n framed submanifolds of S^{n+k} correspond bijectively to based homotopy classes

$$[S^{n+k}, S^n]_* = \pi_{n+k}(S^n)$$

If $f : S^{n+k} \rightarrow S^n$ is based, then $f(\infty) = *$, so for any regular value $q \in S^n \setminus \{*\}$, the regular preimage $f^{-1}(q)$ is automatically disjoint from ∞ . Hence, the framed submanifold associated with f lies in

$$S^{n+k} \setminus \{\infty\} \cong \mathbb{R}^{n+k}$$

Conversely, a compact framed submanifold of $\mathbb{R}^{n+k}$ determines a collapse map on $(\mathbb{R}^{n+k})^+ = S^{n+k}$ which sends ∞ to the base point of S^n . Therefore, the correspondence is exactly the claimed bijection. $\square$

We next describe the suspension homomorphism in this geometric model. Write

$$\mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e$$

where e denotes the unit vector in the last coordinate direction.

Proposition 7.2. *Under the identification of Proposition 7.1, the suspension homomorphism*

$$E : \pi_{n+k}(S^n) \rightarrow \pi_{n+k+1}(S^{n+1})$$

is represented by

$$(M, \varphi) \mapsto (M, \varphi, e)$$

where $M \subseteq \mathbb{R}^{n+k} = \mathbb{R}^{n+k} \times \{0\} \subseteq \mathbb{R}^{n+k+1}$ and (M, φ, e) means that the framing of $\nu_M(\mathbb{R}^{n+k+1})$ is obtained from the framing φ of $\nu_M(\mathbb{R}^{n+k})$ by adjoining the constant normal vector e .

Proof. Let (M, φ) be a compact codimension n framed submanifold of $\mathbb{R}^{n+k}$, and choose a framed tubular neighborhood

$$\Theta : M \times D^n \xrightarrow{\cong} U \subseteq \mathbb{R}^{n+k}$$

The Pontryagin-Thom collapse map is

$$c_{M, \varphi} : (\mathbb{R}^{n+k})^+ \rightarrow D^n / \partial D^n = S^n$$

Now regard M as a submanifold of $\mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e$. Then

$$U \times D^1 \cong M \times D^n \times D^1$$

is a framed tubular neighborhood of M in $\mathbb{R}^{n+k+1}$, and the associated framing is exactly (φ, e) .

The corresponding collapse map is

$$c_{M, \varphi, e} : (\mathbb{R}^{n+k+1})^+ \rightarrow D^{n+1} / \partial D^{n+1} = S^{n+1}$$

Under the canonical identification

$$(\mathbb{R}^{n+k+1})^+ \cong \Sigma(\mathbb{R}^{n+k})^+$$

the map $c_{M,\varphi,e}$ is the suspension of $c_{M,\varphi}$. Therefore,

$$\text{PT}[(M, \varphi, e)] = E(\text{PT}[(M, \varphi)])$$

which proves the claim. $\square$

The proof of the Freudenthal theorem reduces to two geometric propositions.

Proposition 7.3 (Flattening Lemma). *Let $H \cong \mathbb{R}^m$, and let $X^d \subseteq H \oplus \mathbb{R}$ be a compact smooth embedded submanifold. If $\partial X \neq \emptyset$, assume that*

$$\partial X \subseteq H \times \{0\}$$

If $m > 2d$, then there exists an ambient isotopy of $H \oplus \mathbb{R}$ carrying X into $H \times \{0\}$. If $\partial X \neq \emptyset$, the isotopy may be chosen relative to ∂X .

Proof. Write the inclusion of X as

$$j = (g, h) : X \hookrightarrow H \oplus \mathbb{R}$$

where $g : X \rightarrow H$ and $h : X \rightarrow \mathbb{R}$ are smooth maps.

For each $\lambda \in H$, define the shear diffeomorphism

$$S_\lambda : H \oplus \mathbb{R} \rightarrow H \oplus \mathbb{R}, \quad S_\lambda(y, s) = (y + \lambda s, s)$$

Since S_λ fixes $H \times \{0\}$ pointwise, it is already relative to the boundary hyperplane. Define

$$j_\lambda := S_\lambda \circ j = (g_\lambda, h), \quad g_\lambda(x) := g(x) + h(x)\lambda$$

We claim that for some $\lambda \in H$, the map $g_\lambda : X \rightarrow H$ is an embedding.

We first exclude the failure of immersion. Suppose that g_λ is not immersive at some $x \in X$. Then there exists $0 \neq v \in T_x X$ such that

$$(dg)_x(v) + \lambda(dh)_x(v) = 0$$

Since j is an embedding, we have

$$((dg)_x(v), (dh)_x(v)) \neq (0, 0)$$

Hence $(dh)_x(v) \neq 0$, and therefore

$$\lambda = -\frac{(dg)_x(v)}{(dh)_x(v)}$$

Let $S(TX)$ be the unit tangent sphere bundle of X , which has dimension $2d - 1$. On the open set

$$U_{\text{imm}} := \{(x, v) \in S(TX) \mid (dh)_x(v) \neq 0\}$$

define

$$A : U_{\text{imm}} \rightarrow H, \quad A(x, v) = -\frac{(\text{d}g)_x(v)}{(\text{d}h)_x(v)}$$

If λ makes g_λ fail to be an immersion, then $\lambda \in \text{im}(A)$. Since

$$\dim U_{\text{imm}} = 2d - 1 < m = \dim H$$

every point of U_{imm} is a critical point of A , and Sard's theorem implies that $\text{im}(A)$ has measure zero in H .

Next, we exclude the failure of injectivity. Suppose there exist distinct points $x, y \in X$ such that $g_\lambda(x) = g_\lambda(y)$. Then

$$g(x) - g(y) + \lambda(h(x) - h(y)) = 0$$

If $h(x) = h(y)$, then also $g(x) = g(y)$, which contradicts the injectivity of j . Thus $h(x) \neq h(y)$, and

$$\lambda = -\frac{g(x) - g(y)}{h(x) - h(y)}$$

On the open subset

$$U_{\text{inj}} := \{(x, y) \in X \times X \setminus \Delta_X \mid h(x) \neq h(y)\}$$

define

$$B : U_{\text{inj}} \rightarrow H, \quad B(x, y) = -\frac{g(x) - g(y)}{h(x) - h(y)}$$

If λ makes g_λ fail to be injective, then $\lambda \in \text{im}(B)$. Since

$$\dim U_{\text{inj}} = 2d < m = \dim H$$

every point of U_{inj} is a critical point of B , and again Sard's theorem shows that $\text{im}(B)$ has measure zero.

Therefore, the bad set

$$\text{im}(A) \cup \text{im}(B) \subseteq H$$

has measure zero. Choose

$$\lambda \in H \setminus (\text{im}(A) \cup \text{im}(B))$$

Then g_λ is an injective immersion. Since X is compact and H is Hausdorff, g_λ is an embedding.

It follows that $j_\lambda(X)$ is the graph of the smooth function

$$\tilde{h}_0 := h \circ g_\lambda^{-1} : g_\lambda(X) \rightarrow \mathbb{R}$$

Since $g_\lambda(X) \subseteq H$ is a closed embedded submanifold, $\tilde{h}_0$ extends to a smooth function

$$\tilde{h} : H \rightarrow \mathbb{R}$$

If $\partial X \neq \emptyset$, then $\tilde{h}_0 = 0$ on $g_\lambda(\partial X)$ because $\partial X \subseteq H \times \{0\}$, and we may choose the extension so that

$$\tilde{h}|_{g_\lambda(\partial X)} = 0$$

Now define

$$T_t : H \oplus \mathbb{R} \rightarrow H \oplus \mathbb{R}, \quad T_t(y, s) = (y, s - t\tilde{h}(y)), \quad 0 \leq t \leq 1$$

Each T_t is a diffeomorphism, with inverse

$$T_t^{-1}(y, s) = (y, s + t\tilde{h}(y))$$

At $t = 1$, the graph of $\tilde{h}_0$ is sent to $H \times \{0\}$. Since the path $t \mapsto S_{t\lambda}$ is an ambient isotopy from the identity to S_λ , the composition of these two isotopies carries X into $H \times \{0\}$. If $\partial X \neq \emptyset$, then both isotopies fix ∂X : the shear fixes the whole hyperplane $H \times \{0\}$, and T_t fixes $g_\lambda(\partial X) \times \{0\}$ because $\tilde{h} = 0$ there. $\square$

Proposition 7.4 (Splitting Lemma). *Let $X^d \subseteq H \cong \mathbb{R}^{d+c}$ be a compact smooth submanifold, and regard X as a submanifold of $H \oplus \mathbb{R}e$. Let*

$$\Phi = (v_1, \dots, v_{c+1})$$

be an orthonormal framing of the normal bundle $\nu_X(H \oplus \mathbb{R}e)$. If $d < c$, then Φ is homotopic through orthonormal framings to a framing of the form $(\psi_1, \dots, \psi_c, e)$, where $(\psi_1, \dots, \psi_c)$ is an orthonormal framing of $\nu_X(H)$.

If $\partial X \neq \emptyset$ and $v_{c+1} = e$ on ∂X , then the homotopy may be chosen relative to ∂X .

Proof. Since Φ is an orthonormal frame of the normal bundle, the ambient vector e may be written uniquely as

$$e = \sum_{i=1}^{c+1} u_i(x) v_i(x), \quad x \in X$$

where $u = (u_1, \dots, u_{c+1}) : X \rightarrow S^c$ is a smooth map. If $\partial X \neq \emptyset$ and $v_{c+1} = e$ on ∂X , then

$$u|_{\partial X} = e_{c+1} \in S^c$$

Since X is a smooth d -manifold, it has the homotopy type of a CW complex of dimension d . Because $d < c$ and S^c is $(c-1)$ -connected, every map $X \rightarrow S^c$ is null-homotopic. Hence u is homotopic to the constant map e_{c+1} . In the relative case, the same argument applied to the quotient $X/\partial X$ shows that u is homotopic to the constant map e_{c+1} relative to ∂X .

Now consider the Stiefel fibration

$$O(c) \longrightarrow O(c+1) \xrightarrow{p} S^c, \quad p(A) = Ae_{c+1}$$

Choose a homotopy

$$u_t : X \rightarrow S^c, \quad 0 \leq t \leq 1$$

such that

$$u_0 = e_{c+1}, \quad u_1 = u$$

and in the relative case

$$u_t|_{\partial X} = e_{c+1} \quad \forall t \in [0, 1]$$

Since p is a fiber bundle, it has the homotopy lifting property. Therefore, there exists a lift $A_t : X \rightarrow O(c+1)$ such that

$$A_0 = I, \quad A_t e_{c+1} = u_t$$

In the relative case, the lift may be chosen so that

$$A_t|_{\partial X} = I \quad \forall t \in [0, 1]$$

Define a new homotopy of orthonormal framings by

$$\Phi_t := \Phi \cdot A_t^{-1}$$

Here, the matrix A_t^{-1} acts on the ordered frame Φ on the right. At $t = 1$, the coordinates of the ambient vector e with respect to the framing Φ_1 are $A_1^{-1}u = e_{c+1}$, because $u = A_1 e_{c+1}$. Therefore, the last vector of Φ_1 is exactly e .

Since Φ_1 is orthonormal and its last vector is e , its first c vectors are orthogonal to e , hence they lie in the hyperplane $H = e^\perp$. They are also normal to TX , so they form an orthonormal framing of

$$\nu_X(H) = \nu_X(H \oplus \mathbb{R}e) \cap H$$

Thus, $\Phi_1 = (\psi_1, \dots, \psi_c, e)$ for some orthonormal framing $(\psi_1, \dots, \psi_c)$ of $\nu_X(H)$, as required. $\square$

We may now prove the Freudenthal theorem.

Theorem 7.5 (Freudenthal Suspension Theorem for Spheres). *For $n \geq 1$, the suspension homomorphism*

$$E : \pi_q(S^n) \rightarrow \pi_{q+1}(S^{n+1})$$

is an isomorphism for $q \leq 2n - 2$ and a surjection for $q = 2n - 1$.

Proof. Write $q = n + k$. Then

$$q \leq 2n - 1 \iff k < n, \quad q \leq 2n - 2 \iff k + 1 < n.$$

Thus it is enough to prove that

$$E : \pi_{n+k}(S^n) \rightarrow \pi_{n+k+1}(S^{n+1})$$

is surjective when $k < n$ and injective when $k + 1 < n$.

We first prove surjectivity. Let

$$\alpha \in \pi_{n+k+1}(S^{n+1}).$$

By Proposition 7.1, α is represented by a compact codimension $(n + 1)$ framed k -submanifold

$$(M, \Phi) \subseteq \mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e.$$

After applying fiberwise Gram-Schmidt, we may assume that Φ is orthonormal. Since

$$n + k > 2k$$

by the assumption $k < n$, Proposition 7.3 applies with $d = k$ and $m = n + k$. Hence, after an ambient isotopy of $\mathbb{R}^{n+k+1}$, we may assume that $M \subseteq \mathbb{R}^{n+k} \times \{0\}$. Transport the framing Φ along this isotopy.

Now apply Proposition 7.4 with $d = k$ and $c = n$. Since $k < n$, the framing Φ is homotopic to one of the form (φ, e) , where φ is a framing of the codimension n normal bundle of M in $\mathbb{R}^{n+k}$.

Therefore, (M, φ) determines a class $\beta \in \pi_{n+k}(S^n)$. By Proposition 7.2, the suspended class $E(\beta)$ is represented by (M, φ, e) , which is framed cobordant to (M, Φ) . Hence, $E(\beta) = \alpha$. This proves surjectivity when $k < n$.

We now prove injectivity. Suppose $\alpha_0, \alpha_1 \in \pi_{n+k}(S^n)$ satisfy

$$E(\alpha_0) = E(\alpha_1)$$

Choose compact codimension n framed k -submanifolds

$$(M_0, \varphi_0), \quad (M_1, \varphi_1) \subseteq \mathbb{R}^{n+k}$$

representing α_0 and α_1 , and after fiberwise Gram-Schmidt assume that φ_0 and φ_1 are orthonormal. Then $E(\alpha_0)$ and $E(\alpha_1)$ are represented by (M_0, φ_0, e) and (M_1, φ_1, e) inside $\mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e$.

Since these two classes are equal, there exists a framed cobordism

$$(W, \Phi) \subseteq (\mathbb{R}^{n+k+1}) \times [0, 1]$$

from (M_0, φ_0, e) to (M_1, φ_1, e) . By orthonormalizing Φ fiberwise and keeping the boundary framing fixed, we may assume that Φ is orthonormal and that on the boundary the last normal vector is e .

Rewrite the ambient space as

$$(\mathbb{R}^{n+k} \times [0, 1]) \oplus \mathbb{R}e$$

Since $\dim W = k + 1$ and $\dim (\mathbb{R}^{n+k} \times [0, 1]) = n + k + 1$, the inequality $k + 1 < n$ implies

$$n + k + 1 > 2(k + 1)$$

Hence Proposition 7.3 applies, relative to the boundary, and carries W into

$$\mathbb{R}^{n+k} \times [0, 1]$$

Transport the framing along this isotopy.

Now apply the relative form of Proposition 7.4 with $d = k + 1$ and $c = n$. Since $k + 1 < n$, the framing Φ is homotopic rel boundary to a framing of the form (Ψ, e) , where Ψ is a framing of the codimension n normal bundle of W in $\mathbb{R}^{n+k} \times [0, 1]$.

Therefore, (W, Ψ) is a framed cobordism in $\mathbb{R}^{n+k} \times [0, 1]$ from (M_0, φ_0) to (M_1, φ_1) . By Proposition 7.1, this means $\alpha_0 = \alpha_1$. Hence E is injective when $k + 1 < n$.

Combining the two parts, E is an isomorphism for $q \leq 2n - 2$ and a surjection for $q = 2n - 1$. $\square$

A standard consequence is the stability of the homotopy groups $\pi_{n+k}(S^n)$ once n is sufficiently large relative to k .

Corollary 7.6. *Fix $k \geq 0$. Then the suspension map*

$$E : \pi_{n+k}(S^n) \rightarrow \pi_{n+k+1}(S^{n+1})$$

is an isomorphism for every $n > k + 1$. Consequently, the groups $\pi_{n+k}(S^n)$ are independent of n in this range.

Proof. If $n > k + 1$, then $k + 1 < n$, so Theorem 7.5 implies that the suspension map is an isomorphism. Repeating suspension, therefore, identifies all the groups in the stable range. $\square$

7.2 Calculation of $\pi_{n+1}(S^n)$

We first calculate the first unstable-looking group that becomes stable after one suspension. The answer is that, except for the Hopf fibration case $n = 2$, the group $\pi_{n+1}(S^n)$ is $\mathbb{Z}/2$.

The Pontryagin-Thom construction gives the geometric interpretation

$$\pi_{n+1}(S^n) = [S^{n+1}, S^n]$$

If $q \in S^n$ is a regular value of a smooth map $f : S^{n+1} \rightarrow S^n$, then $f^{-1}(q)$ is a closed 1-dimensional framed submanifold of S^{n+1} .

Thus, for $n \geq 3$, the problem is equivalent to classifying closed framed 1-manifolds up to framed cobordism. The codimension is at least 3, so the framed cobordism group is already stable. We denote it by Ω_1^{fr} .

Theorem 7.7. *The framed bordism group of closed 1-manifolds is $\Omega_1^{\text{fr}} \cong \mathbb{Z}/2$.*

Proof. Let (L, φ) be a closed framed 1-manifold. Since every closed 1-manifold is a disjoint union of circles, we may write

$$L = S_1^1 \sqcup \cdots \sqcup S_k^1$$

A normal framing of L determines a stable framing of TL . Indeed, for a framed submanifold $L \subseteq S^{n+1}$, the stable triviality

$$TS^{n+1} \oplus \mathbb{R} \cong \mathbb{R}^{n+2}$$

and the normal framing $\nu_L(S^{n+1}) \cong L \times \mathbb{R}^n$ imply

$$TL \oplus \mathbb{R}^n \oplus \mathbb{R} \cong L \times \mathbb{R}^{n+2}$$

Hence, each component S_i^1 obtains a stable framing.

There are exactly two stable framings of S^1 up to homotopy. One extends over a disk D^2 , and the other does not. Define

$$\epsilon(S_i^1, \varphi_i) = \begin{cases} 0, & \text{if the induced stable framing extends over } D^2, \\ 1, & \text{otherwise} \end{cases}$$

Then define

$$\alpha(L, \varphi) := \sum_{i=1}^k \epsilon(S_i^1, \varphi_i) \in \mathbb{Z}/2$$

This number is invariant under framed cobordism. Indeed, if (L_0, φ_0) and (L_1, φ_1) are framed cobordant by a framed surface W , then the framing of W gives a stable framing of TW . Therefore, the boundary stable framings on L_0 and L_1 differ by a stable framed cobordism. The obstruction to extending the boundary stable framing over W has zero total boundary contribution, and hence

$$\alpha(L_0, \varphi_0) = \alpha(L_1, \varphi_1)$$

The invariant α is surjective because a circle with the non-bounding stable framing has $\alpha = 1$. It remains to show that α is complete.

If $\alpha(L, \varphi) = 0$, then the number of components with non-bounding stable framing is even. A component with bounding framing bounds a framed disk. Two components with non-bounding framings together bound a framed annulus. Therefore, all components of L can be killed by framed cobordisms, and (L, φ) is framed null-cobordant.

Thus, the framed cobordism class is completely determined by $\alpha \in \mathbb{Z}/2$. Hence, $\Omega_1^{\text{fr}} \cong \mathbb{Z}/2$. $\square$

Under Pontryagin-Thom, the nonzero element of Ω_1^{fr} corresponds to the suspension of the Hopf map. Let $\eta_2 : S^3 \rightarrow S^2$ be the Hopf fibration. For $n \geq 3$, define

$$\eta_n := \Sigma^{n-2} \eta_2 \in \pi_{n+1}(S^n)$$

This element is represented by a framed circle with the non-bounding stable framing.

Theorem 7.8. *For $n \geq 3$, $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$. The nonzero element is represented by η_n , and $2\eta_n = 0$.*

Proof. By the Pontryagin-Thom construction, a class in $\pi_{n+1}(S^n)$ is represented by a closed codimension n framed 1-submanifold of S^{n+1} . Since $n \geq 3$, this framed cobordism group is the stable framed bordism group Ω_1^{fr} .

By Theorem 7.7, $\Omega_1^{\text{fr}} \cong \mathbb{Z}/2$. Therefore

$$\pi_{n+1}(S^n) \cong \mathbb{Z}/2$$

The generator is the class represented by a framed circle whose induced stable framing is non-bounding. Under Pontryagin-Thom, this is exactly the suspended Hopf class η_n . Since the group is $\mathbb{Z}/2$, we have $2\eta_n = 0$. $\square$

For completeness, the low-dimensional cases are

$$\pi_2(S^1) = 0, \quad \pi_3(S^2) \cong \mathbb{Z}$$

The second group is generated by the Hopf fibration $\eta_2 : S^3 \rightarrow S^2$. Thus the stable answer $\mathbb{Z}/2$ begins at $n = 3$.

Therefore,

$$\pi_{n+1}(S^n) \cong \begin{cases} 0, & n = 1, \\ \mathbb{Z}, & n = 2, \\ \mathbb{Z}/2, & n \geq 3 \end{cases}$$

7.3 Calculation of $\pi_{n+2}(S^n)$

We now calculate $\pi_{n+2}(S^n)$. For $n \geq 4$, the Pontryagin-Thom construction identifies this group with the framed bordism group of closed framed surfaces:

$$\pi_{n+2}(S^n) \cong \Omega_2^{\text{fr}}$$

The answer is again $\mathbb{Z}/2$, but the invariant is subtler than in dimension 1. It is the Arf invariant of the spin structure determined by the framing.

Let (Σ, φ) be a closed framed surface. The normal framing and the stable triviality of the sphere determine a stable framing of $T\Sigma$, and hence a spin structure on Σ . A spin structure on an oriented closed surface determines a quadratic refinement $q_\varphi : H_1(\Sigma; \mathbb{F}_2) \rightarrow \mathbb{F}_2$ of the mod-2 intersection pairing. This means

$$q_\varphi(a + b) = q_\varphi(a) + q_\varphi(b) + a \cdot b$$

for all $a, b \in H_1(\Sigma; \mathbb{F}_2)$.

Choose a symplectic basis $a_1, b_1, \dots, a_g, b_g$ of $H_1(\Sigma; \mathbb{F}_2)$, so that $a_i \cdot b_i = 1$ and all other intersections vanish. The Arf invariant of q_φ is

$$\text{Arf}(q_\varphi) = \sum_{i=1}^g q_\varphi(a_i)q_\varphi(b_i) \in \mathbb{F}_2$$

This number is independent of the chosen symplectic basis.

Theorem 7.9. *The framed bordism group of closed surfaces is*

$$\Omega_2^{\text{fr}} \cong \mathbb{Z}/2$$

The isomorphism is given by the Arf invariant.

Proof. We define $\text{Arf}(\Sigma, \varphi) := \text{Arf}(q_\varphi) \in \mathbb{F}_2$. First, this is invariant under framed cobordism. If (Σ_0, φ_0) and (Σ_1, φ_1) are framed cobordant, then there exists a framed 3-manifold W with

$$\partial W = \Sigma_0 \sqcup \Sigma_1$$

The framing of W induces a spin cobordism between the spin structures on Σ_0 and Σ_1 . The Arf invariant is invariant under spin cobordism of surfaces. Hence

$$\text{Arf}(\Sigma_0, \varphi_0) = \text{Arf}(\Sigma_1, \varphi_1)$$

We next show that the Arf invariant is complete. Let (Σ, φ) be a closed framed surface. If there exists a simple closed curve $\gamma \subseteq \Sigma$ with $q_\varphi([\gamma]) = 0$, then the framing along γ is the bounding one. Hence, one can perform framed surgery along γ . This surgery removes a handle and preserves the framed bordism class.

By repeating framed surgeries, we can reduce Σ to a standard form determined only by the Arf invariant. If $\text{Arf}(q_\varphi) = 0$, then one can choose enough curves with $q_\varphi = 0$ to kill all handles. The result is a framed 2-sphere. Since $\pi_2(SO) = 0$, the stable framing on this 2-sphere extends over a 3-ball. Hence, the original framed surface is framed null-cobordant.

If $\text{Arf}(q_\varphi) = 1$, then framed surgeries reduce (Σ, φ) to a framed torus (T^2, φ_T) with a symplectic basis a, b satisfying $q_{\varphi_T}(a) = q_{\varphi_T}(b) = 1$. This torus has Arf invariant 1, and hence it is not framed null-cobordant.

Therefore, every framed surface is framed cobordant either to the empty surface or to this framed torus. Since the Arf invariant is additive under disjoint union, two copies of the framed torus have Arf invariant 0 and are framed null-cobordant. Thus, the framed bordism group is $\mathbb{Z}/2$. $\square$

The nonzero element of Ω_2^{fr} corresponds under Pontryagin-Thom to the stable class usually denoted by η^2 . Here η is the stable Hopf element, and η^2 is its product in the stable homotopy ring.

Theorem 7.10. For $n \geq 4$,

$$\pi_{n+2}(S^n) \cong \mathbb{Z}/2$$

The nonzero element is represented by η_n^2 .

Proof. Let $f : S^{n+2} \rightarrow S^n$ be a smooth representative of a homotopy class. For a regular value $q \in S^n$, the preimage $\Sigma = f^{-1}(q)$ is a closed 2-dimensional framed submanifold of S^{n+2} . Since $n \geq 4$, the codimension is large enough for the framed cobordism group to be stable. Thus Pontryagin-Thom gives $\pi_{n+2}(S^n) \cong \Omega_2^{\text{fr}}$. By Theorem 7.9, $\Omega_2^{\text{fr}} \cong \mathbb{Z}/2$. Hence

$$\pi_{n+2}(S^n) \cong \mathbb{Z}/2.$$

The generator is represented by a framed torus with Arf invariant 1. Under Pontryagin-Thom, this generator is the element η_n^2 . $\square$

It remains to record the two low-dimensional cases $n = 2$ and $n = 3$. They have the same answer, although they do not lie in the stable range used above.

First, consider the Hopf fibration $S^1 \hookrightarrow S^3 \rightarrow S^2$. The long exact sequence in homotopy gives an isomorphism

$$\pi_4(S^3) \cong \pi_4(S^2)$$

because $\pi_4(S^1) = 0$ and $\pi_3(S^1) = 0$. By Theorem 7.8, applied to S^3 , we have $\pi_4(S^3) \cong \mathbb{Z}/2$. Therefore $\pi_4(S^2) \cong \mathbb{Z}/2$.

Next, consider the quaternionic Hopf fibration $S^3 \hookrightarrow S^7 \rightarrow S^4$. The long exact sequence gives

$$\cdots \longrightarrow \pi_6(S^7) \longrightarrow \pi_6(S^4) \longrightarrow \pi_5(S^3) \longrightarrow \pi_5(S^7) \longrightarrow \cdots$$

Since $\pi_6(S^7) = 0$ and $\pi_5(S^7) = 0$, we get an isomorphism

$$\pi_6(S^4) \cong \pi_5(S^3)$$

By Theorem 7.10, applied to $n = 4$, we have $\pi_6(S^4) \cong \mathbb{Z}/2$. Hence $\pi_5(S^3) \cong \mathbb{Z}/2$.

Therefore, for every $n \geq 2$,

$$\pi_{n+2}(S^n) \cong \mathbb{Z}/2$$

Geometrically, this means that a class in $\pi_{n+2}(S^n)$ is detected by a framed surface, and the only obstruction to being framed null-cobordant is the Arf invariant. Thus, the two possible classes are represented by

$$\text{Arf} = 0 \quad \text{and} \quad \text{Arf} = 1.$$

The class with $\text{Arf} = 1$ is the element η^2 .

7.4 J -Homomorphism

Too tired to write this section. I will finish this before next semester :(

7.5 Degree, Vector Fields, and the Poincaré-Hopf Theorem

The purpose of this subsection is to explain the Poincaré-Hopf theorem from the viewpoint of degree theory, zero loci of sections, and cobordism. The basic philosophy is that the degree of a map counts regular preimages as oriented zero-dimensional manifolds, while the index of a vector field counts zeros of a section of the tangent bundle by local degree.

Throughout this subsection, unless otherwise stated, M^n denotes a closed oriented smooth manifold.

Definition 7.11 (Degree via Regular Values). Let M^n and N^n be closed oriented smooth manifolds, and suppose that N is connected. Let $f : M \rightarrow N$ be a smooth map. If $y \in N$ is a regular value of f , then $f^{-1}(y)$ is a finite set. For each $x \in f^{-1}(y)$, the differential

$$(df)_x : T_x M \rightarrow T_y N$$

is a linear isomorphism. Define

$$\epsilon_x(f) = \begin{cases} +1, & (df)_x \text{ preserves orientation,} \\ -1, & (df)_x \text{ reverses orientation} \end{cases}$$

The degree of f is defined by

$$\deg(f) := \sum_{x \in f^{-1}(y)} \epsilon_x(f)$$

Proposition 7.12 (Degree as a Zero-Dimensional Cobordism Invariant). *Let $f_0, f_1 : M^n \rightarrow N^n$ be smooth maps between closed oriented smooth manifolds, and suppose that f_0 and f_1 are smoothly homotopic. Then $\deg(f_0) = \deg(f_1)$.*

Proof. Let $F : M \times [0, 1] \rightarrow N$ be a smooth homotopy from f_0 to f_1 , i.e.

$$F(x, 0) = f_0(x), \quad F(x, 1) = f_1(x)$$

By a small perturbation of F relative to the boundary if necessary, we may assume that a chosen regular value $y \in N$ is regular for F , f_0 , and f_1 .

Consider $W := F^{-1}(y)$. Since y is a regular value of F , the preimage theorem gives that W is a compact oriented 1-dimensional manifold. Moreover, its boundary is

$$\partial W = f_1^{-1}(y) - f_0^{-1}(y)$$

where the minus sign comes from the boundary orientation on $M \times [0, 1]$.

A compact oriented 1-manifold is a finite disjoint union of circles and compact intervals. Every compact interval has two boundary points with opposite signs. Hence, the algebraic boundary count of W is zero. Therefore

$$\sum_{x \in f_1^{-1}(y)} \epsilon_x(f_1) - \sum_{x \in f_0^{-1}(y)} \epsilon_x(f_0) = 0$$

Thus, $\deg(f_1) = \deg(f_0)$. □

Remark 7.13. This proof shows that the degree is fundamentally a cobordism invariant. The regular preimage $f^{-1}(y)$ is an oriented zero-dimensional manifold, and the homotopy F produces an oriented cobordism between the two zero-dimensional manifolds

$$f_0^{-1}(y) \quad \text{and} \quad f_1^{-1}(y)$$

Since $\Omega_0^{SO} \cong \mathbb{Z}$, the cobordism class of an oriented finite set is exactly its signed count.

Definition 7.14 (Local Degree). Let $U \subseteq \mathbb{R}^n$ be open, and let $g : U \rightarrow \mathbb{R}^n$ be a smooth map. Suppose that $a \in U$ is an isolated zero of g , i.e. $g(a) = 0$, $g(x) \neq 0$ for $0 < \|x - a\| \ll 1$. Choose $\varepsilon > 0$ sufficiently small such that $\overline{B_\varepsilon(a)} \subseteq U$ and $g(x) \neq 0$ for $x \in \partial B_\varepsilon(a)$. The local degree of g at a is defined by

$$\deg_a(g) := \deg\left(\frac{g}{\|g\|} : \partial B_\varepsilon(a) \rightarrow S^{n-1}\right)$$

Proposition 7.15 (Local Degree at a Nondegenerate Zero). Let $g : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ be smooth, and suppose that $a \in U$ is a nondegenerate zero, i.e.

$$g(a) = 0, \quad Dg_a : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ is an isomorphism}$$

Then, $\deg_a(g) = \text{sgn det}(Dg_a)$.

Proof. Since Dg_a is invertible, the inverse function theorem implies that a is an isolated zero. For sufficiently small $\varepsilon > 0$, the map g on $\partial B_\varepsilon(a)$ is homotopic through non-vanishing maps to its linear approximation $x \mapsto Dg_a(x - a)$. Hence,

$$\deg_a(g) = \deg\left(\frac{Dg_a(x - a)}{\|Dg_a(x - a)\|} : \partial B_\varepsilon(a) \rightarrow S^{n-1}\right)$$

The last degree is $+1$ if Dg_a preserves orientation and -1 if Dg_a reverses orientation. Therefore, $\deg_a(g) = \text{sgn det}(Dg_a)$, which proves the proposition. $\square$

Definition 7.16 (Index of a Vector Field). Let $X \in \Gamma(TM)$ be a smooth vector field on an oriented smooth manifold M^n , and let $p \in M$ be an isolated zero of X . Choose an oriented coordinate chart

$$(U, \varphi) = (U, x^1, \dots, x^n)$$

around p such that $\varphi(p) = 0$. Write

$$X|_U = \sum_{i=1}^n X^i \frac{\partial}{\partial x^i}$$

This gives a smooth map

$$X_\varphi : \varphi(U) \rightarrow \mathbb{R}^n, \quad X_\varphi = (X^1, \dots, X^n)$$

The index of X at p is defined by $\text{ind}_p(X) := \deg_0(X_\varphi)$.

Proposition 7.17 (Well-Definedness and Nondegenerate Formula for the Index). The integer $\text{ind}_p(X)$ is independent of the chosen oriented coordinate chart. Moreover, if p is a nondegenerate zero of X , then

$$\text{ind}_p(X) = \text{sgn det}(\nabla X)_p,$$

where ∇ is any connection on TM .

Proof. First consider the independence of coordinates. If two oriented charts are chosen, then the corresponding local maps differ by composition with orientation-preserving local diffeomorphisms on the source and target. Since the degree is invariant under orientation-preserving changes of coordinates, the local degree is unchanged. Hence $\text{ind}_p(X)$ is well-defined.

Now suppose that p is a nondegenerate zero. Since $X(p) = 0$, the linear map

$$(\nabla X)_p : T_p M \rightarrow T_p M$$

is independent of the choice of connection. In a local coordinate trivialization, this linear map is represented by the Jacobian matrix of the local map X_φ at $\varphi(p) = 0$. Therefore, by the previous proposition, $\text{ind}_p(X) = \text{sgn det}(\nabla X)_p$. $\square$

Proposition 7.18 (Zero Locus of a Transverse Section). *Let $E \rightarrow M^m$ be a smooth rank r vector bundle, and let $s \in \Gamma(E)$ be a smooth section transverse to the zero section. Then $Z(s) := s^{-1}(0)$ is a smooth embedded submanifold of M of dimension*

$$\dim Z(s) = m - r$$

Moreover, for every $p \in Z(s)$, there is a short exact sequence

$$0 \longrightarrow T_p Z(s) \hookrightarrow T_p M \xrightarrow{D_p s} E_p \longrightarrow 0$$

and hence a canonical normal bundle isomorphism $\nu_{Z(s)}(M) \cong E|_{Z(s)}$.

Proof. Locally trivialize E over an open set $U \subseteq M$: $E|_U \cong U \times \mathbb{R}^r$. Under this trivialization, the section s is represented by a smooth map $s_U : U \rightarrow \mathbb{R}^r$. The transversality of s to the zero section is exactly the statement that $0 \in \mathbb{R}^r$ is a regular value of s_U . Therefore, by the preimage theorem (Theorem 4.2), $Z(s) \cap U = s_U^{-1}(0)$ is a smooth submanifold of U of dimension $m - r$.

At $p \in Z(s)$, the tangent space is

$$T_p Z(s) = \ker(D_p s)$$

where $D_p s : T_p M \rightarrow E_p$ is the vertical derivative of the section at the zero point. Since s is transverse to the zero section, $D_p s$ is surjective. Hence, we obtain the short exact sequence

$$0 \longrightarrow T_p Z(s) \hookrightarrow T_p M \xrightarrow{D_p s} E_p \longrightarrow 0$$

Taking the quotient by $T_p Z(s)$ gives $T_p M / T_p Z(s) \cong E_p$. Therefore, $\nu_{Z(s)}(M) \cong E|_{Z(s)}$. $\square$

Remark 7.19. If E is equipped with a framing $E \cong M \times \mathbb{R}^r$. Then, the canonical isomorphism

$$\nu_{Z(s)}(M) \cong E|_{Z(s)}$$

gives a framing of $Z(s) \subseteq M$. This is the basic geometric mechanism behind the Pontryagin-Thom construction. For the Poincaré-Hopf theorem, however, the relevant bundle is $E = TM$, which is not necessarily trivial. Therefore the correct object is not necessarily a framed zero locus, but a zero locus carrying the normal data coming from TM .

Definition 7.20 (Algebraic Zero Count). Let $E \rightarrow M^n$ be an oriented rank n vector bundle over a closed oriented smooth manifold M^n . Let $s \in \Gamma(E)$ be transverse to the zero section. Then $Z(s)$ is a finite set.

For $p \in Z(s)$, define

$$\epsilon_p(s) = \begin{cases} +1, & D_p s : T_p M \rightarrow E_p \text{ preserves orientation,} \\ -1, & D_p s : T_p M \rightarrow E_p \text{ reverses orientation.} \end{cases}$$

The algebraic zero count of s is

$$\#^{\text{alg}} Z(s) := \sum_{p \in Z(s)} \epsilon_p(s)$$

Proposition 7.21 (Cobordism Invariance of the Algebraic Zero Count). Let $E \rightarrow M^n$ be an oriented rank n vector bundle over a closed oriented smooth manifold. If $s_0, s_1 \in \Gamma(E)$ are transverse to the zero section, then

$$\#^{\text{alg}} Z(s_0) = \#^{\text{alg}} Z(s_1)$$

Proof. Choose a smooth homotopy of sections $s_t \in \Gamma(E)$ with $t \in [0, 1]$, from s_0 to s_1 . Equivalently, define a section $S \in \Gamma(\pi_M^* E)$ over $M \times [0, 1]$ by $S(x, t) = s_t(x)$. By a small perturbation relative to the boundary, we may assume that S is transverse to the zero section of $\pi_M^* E$.

Then, $W := Z(S) = S^{-1}(0)$ is a compact oriented 1-dimensional manifold. Its boundary is $\partial W = Z(s_1) - Z(s_0)$. Since every compact oriented 1-manifold has algebraic boundary count zero, we have

$$\#^{\text{alg}} Z(s_1) - \#^{\text{alg}} Z(s_0) = 0$$

Therefore, $\#^{\text{alg}} Z(s_0) = \#^{\text{alg}} Z(s_1)$. □

Definition 7.22 (Euler Number via Zero Loci). Let $E \rightarrow M^n$ be an oriented rank n vector bundle over a closed oriented smooth manifold M^n . The Euler number of E is defined by

$$\langle e(E), [M] \rangle := \#^{\text{alg}} Z(s),$$

where $s \in \Gamma(E)$ is any section transverse to the zero section.

Remark 7.23. The previous proposition proves that the definition is independent of the chosen transverse section. In cohomological language, the zero locus of a transverse section represents the Poincaré dual of the Euler class:

$$[Z(s)] = \text{PD}(e(E))$$

When $\text{rank } E = \dim M$, this becomes the numerical identity

$$\#^{\text{alg}} Z(s) = \langle e(E), [M] \rangle$$

Proposition 7.24 (Vector Fields as Sections of the Tangent Bundle). Let $X \in \Gamma(TM)$ be a vector field on a closed oriented smooth manifold M^n . Suppose that X is transverse to the zero section. Then $Z(X)$ is finite, all zeros are nondegenerate, and

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \langle e(TM), [M] \rangle$$

Proof. A vector field is precisely a section of the tangent bundle $X \in \Gamma(TM)$. Since X is transverse to the zero section, the zero locus $Z(X) = X^{-1}(0)$ is a zero-dimensional submanifold of M , hence a finite set.

At every zero $p \in Z(X)$, the vertical derivative $D_p X : T_p M \rightarrow T_p M$ is an isomorphism. Hence p is a nondegenerate zero. By the nondegenerate index formula, $\text{ind}_p(X) = \text{sgn det}(D_p X)$. This is exactly the sign $\epsilon_p(X)$ appearing in the algebraic zero count of the section X . Therefore

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \#^{\text{alg}} Z(X)$$

By the definition of the Euler number, $\#^{\text{alg}} Z(X) = \langle e(TM), [M] \rangle$. Thus

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \langle e(TM), [M] \rangle$$

which completes the proof. $\square$

Proposition 7.25 (Additivity of the Local Index). *Let X be a vector field on M , and let $p \in M$ be an isolated zero. Choose a coordinate ball B around p such that X has no zero on ∂B . Suppose that X' is another vector field on B such that $X' = X$ near ∂B and suppose that X' has only nondegenerate zeros in B . Then*

$$\text{ind}_p(X) = \sum_{q \in Z(X') \cap B} \text{ind}_q(X')$$

Proof. In a local trivialization over B , the vector fields X and X' are represented by maps $B \rightarrow \mathbb{R}^n$. Since $X' = X$ near ∂B , the normalized maps $X/\|X\|$ and $X'/\|X'\|$ agree on ∂B . Therefore, their degrees on ∂B are the same.

Let $q_1, \dots, q_k$ be the zeros of X' in B . Choose pairwise disjoint small balls B_i around q_i . On the compact manifold with boundary

$$B \setminus \bigcup_{i=1}^k \text{int}(B_i),$$

the vector field X' is nowhere zero. Hence the normalized map

$$\frac{X'}{\|X'\|} : B \setminus \bigcup_{i=1}^k \text{int}(B_i) \rightarrow S^{n-1}$$

gives a cobordism between the outer boundary map on ∂B and the inner boundary maps on the ∂B_i . Therefore the degree on the outer boundary equals the sum of the degrees on the inner boundaries:

$$\deg \left(\frac{X'}{\|X'\|} : \partial B \rightarrow S^{n-1} \right) = \sum_{i=1}^k \deg \left(\frac{X'}{\|X'\|} : \partial B_i \rightarrow S^{n-1} \right)$$

The left-hand side is $\text{ind}_p(X)$, while the i -th term on the right-hand side is $\text{ind}_{q_i}(X')$. Hence

$$\text{ind}_p(X) = \sum_{q \in Z(X') \cap B} \text{ind}_q(X')$$

which completes the proof. $\square$

Theorem 7.26 (Poincaré-Hopf Theorem). *Let M^n be a closed oriented smooth manifold, and let $X \in \Gamma(TM)$ be a smooth vector field with isolated zeros. Then*

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \chi(M).$$

Proof. We first prove the theorem for vector fields transverse to the zero section. If $X \pitchfork 0_M$, then by the previous proposition,

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \langle e(TM), [M] \rangle.$$

Thus, it remains to compute the Euler number of the tangent bundle.

Choose a Morse function $f : M \rightarrow \mathbb{R}$ and a Riemannian metric g on M . Let $Y = \nabla f$ be the gradient vector field. The zeros of Y are precisely the critical points of f : $Z(Y) = \text{Crit}(f)$. Since f is Morse, every critical point is nondegenerate, so Y is transverse to the zero section.

Let $p \in \text{Crit}(f)$, and let $\lambda(p)$ be the Morse index of p . At p , the linearization of ∇f is given by

$$D_p(\nabla f) = g_p^{-1} \circ \text{Hess}_p(f).$$

Since g_p^{-1} is positive definite, it has a positive determinant. Therefore, the sign of $\det D_p(\nabla f)$ is the same as the sign of the determinant of the Hessian of f at p .

By the Morse lemma, the Hessian of f at p has $\lambda(p)$ negative eigenvalues and $n - \lambda(p)$ positive eigenvalues. Hence, $\text{ind}_p(\nabla f) = (-1)^{\lambda(p)}$. Therefore,

$$\sum_{p \in Z(\nabla f)} \text{ind}_p(\nabla f) = \sum_{p \in \text{Crit}(f)} (-1)^{\lambda(p)}.$$

Let c_k be the number of critical points of f of Morse index k . Then

$$\sum_{p \in \text{Crit}(f)} (-1)^{\lambda(p)} = \sum_{k=0}^n (-1)^k c_k.$$

By Morse theory, M admits a CW decomposition with exactly c_k cells of dimension k . Hence

$$\chi(M) = \sum_{k=0}^n (-1)^k c_k.$$

Consequently,

$$\langle e(TM), [M] \rangle = \sum_{p \in Z(\nabla f)} \text{ind}_p(\nabla f) = \chi(M).$$

Thus, for every vector field transverse to the zero section,

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \chi(M).$$

Now let X be any vector field with isolated zeros. Choose pairwise disjoint coordinate balls B_p for $p \in Z(X)$ such that X has no zero on ∂B_p . Perturb X inside the balls B_p , keeping it fixed near each

∂B_p , to obtain a vector field X' whose zeros are all nondegenerate. By the additivity of the local index,

$$\text{ind}_p(X) = \sum_{q \in Z(X') \cap B_p} \text{ind}_q(X').$$

Summing over all $p \in Z(X)$ gives

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \sum_{q \in Z(X')} \text{ind}_q(X')$$

Since X' is transverse to the zero section, the first part of the proof gives

$$\sum_{q \in Z(X')} \text{ind}_q(X') = \chi(M)$$

Therefore

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \chi(M)$$

which completes the proof of Poincaré-Hopf Theorem, □

Remark 7.27. The cobordism appearing in the Poincaré-Hopf theorem is not a cobordism of the ambient manifolds. In particular, the Euler characteristic is not an oriented cobordism invariant of manifolds. Rather, the cobordism is the cobordism of zero loci of homotopic sections. For vector fields, these sections are sections of TM , and their zero loci represent the Euler class of TM .

References

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